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Surrogate-based real-time curbside management for ride-hailing and delivery operations

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ABSTRACT

This work investigates surrogate model-based optimization for real-time curbside traffic management to minimize congestion caused by the stopping of vehicles along the curbside (e.g. for ride-hailing or delivery operations). A hybrid simulation approach where the main traffic flows interact with individually modelled stopping vehicles can be adopted to reproduce the objective of the optimization problem due to its non-linearity and coupled with any meta-heuristic. However, because simulations are time expensive and hence unsuitable for real-time control, a surrogate model that takes the decision variables as inputs and approximates the objective function is employed to replace the simulation within the solution algorithm. Several modelling techniques are compared by their accuracy and computational tractability for real-time control under various scenarios. Finally, a realistic application with multiple pick-up/drop-off operations is presented showing the potential of the proposed approach in reducing the negative impacts of stopping vehicles and favorable computational properties.

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KEYWORDS
Traffic optimization; curbside management; surrogate modelling; simulation-based optimization

1. Introduction

Due to recent changes in mobility, such as the growth of ride-hailing and delivery services, urban curbside is facing increasing demand (Marsden, Docherty, and Dowling 2020). In urban areas already burdened by limited road capacity and parking, it is crucial to find new solutions to maximize the efficiency of these new operations. The advancements in wireless communication, computing, and sensor technology over the past decade have paved the way for new solutions based on Vehicle-to-Infrastructure (V2I) communication systems. Sensors can monitor parking availability and traffic in real-time, sharing data with equipped automobiles over 5G networks (Al-Turjman and Malekloo 2019). New traffic management methods for urban delivery vehicles can be developed using these technologies to balance curbside utilization and reduce road congestion. This especially applies to high-demand locations like theaters, sports facilities, and transport hubs, where the lack of dedicated drop-off points leads to vehicles blocking traffic and worsening congestion. Real-time traffic management can reduce congestion by dynamically affecting vehicles' movements through advanced sensing and communication technologies (Rizwan, Suresh, and Babu 2016). For example, in the specific case of curbside, it can provide directions for curbside stop positions on a street or network to minimize traffic disruptions. This strategy can be formulated as a dynamic optimization (or control) problem in which decision variables (e.g. signal, curbside allocations) are determined based on time-varying information on traffic conditions and curbside demand.

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Given the above requirements, in the present work, we propose a novel optimization formulation for controlling the stop positions of vehicles intending to make curbside stops on a multi-lane road segment to perform deliveries, pick-up, and drop-off operations. The goal of the optimization is to reduce the adverse effects of the stops on the traffic flow in the segment. A surrogate-modelling-based solution approach to the problem is proposed, analyzed, and demonstrated through an example application in allocating pick-up and drop-off operations for ride-hailing service vehicles.

Simulation-based optimization (SO) can be employed to solve control problems. Within this framework, simulation is adopted to evaluate the objective function in relation to the decision variables and integrated with a generic optimization approach to identify well-performing solutions. Memetic algorithms, simulated annealing, and genetic algorithms (GAs) are all examples of evolutionary algorithms that can be straightforwardly coupled to optimization for traffic management purposes (Liard, Stern, and Delle Monache 2023). However, depending on the nature of the problem, evolutionary algorithms typically require a large number of simulation assessments and, ultimately, a significant computational effort to reach high-quality solutions. Therefore, it is challenging to implement these solutions for real-time control problems.

Surrogate modelling creates a statistical model (called surrogate model) to represent the simulation's output and can be employed to obviate simulation. A trained surrogate model can be integrated with optimization in place of the original simulation for several purposes such as discussed in Section 2. This paper develops alternative surrogate model-based optimization approaches for managing road curbside to maximize its traffic throughput and analyzes their accuracy and computational efficiency. The studied problem is formulated as a control problem that can be solved through a model predictive control (MPC) framework (Kouvaritakis and Cannon 2016). The present paper introduces novel research on optimizing the positions of vehicle stoppages while considering traffic dynamics by means of surrogate modelling, which, to the best of the authors' knowledge, has not been previously studied. The contributions of the paper are summarized as follows:

(1) We present a novel optimization problem to control the curbside vehicle stop positions with the objective of minimizing the adverse effects of the congestion caused by the stops on the traffic flow. The problem is implemented within an MPC framework based on a macroscopic traffic model solved with a Lax–Hopf-based simulation scheme.
(2) We investigate a surrogate-based solution approach for solving the given optimization problem considering various alternatives of the surrogate model. A numerical analysis is performed to determine the best surrogate model, considering solution accuracy and computation time.
(3) We present an illustrative application of the proposed control approach in the context of determining stop positions of pick-up and drop-off operations in ride-hailing services. The control approach improves the traffic performance objective by 20.65% and 17.94% using direct simulation and surrogate models, respectively, without detour penalties for stop position deviations, by reducing upstream spillback from vehicle stoppages and increasing the traffic outflow.

The rest of the paper is organized as follows- Section 2 presents a review of the literature related to the current work. Section 3 presents the traffic control problem along with a description of the traffic dynamics and the optimization problem. The approach to solve the said optimization problem using a surrogate model is described in Section 4 along with a comparative analysis of the various surrogate models used. Finally, a numerical example of the application of the proposed control approach is presented in Section 5 followed by a conclusion in Section 6.

2. Related work

Several studies have tackled the issue of logistics and ride-hailing operations' traffic impacts and related curbside management by employing traffic simulation. Roca-Riu et al. (2017) propose 'Dynamic

Delivery Parking Spots'- delivery facilities positioned on a link's shoulder lane that are activated to minimize traffic delays. The proposed analytical model is validated with simulation. Ukkusuri et al. (2016) assess the impact of off-hour delivery operations on traffic flow with the help of a regional travel demand model and a mesoscopic traffic simulation model. Stueger, Fehn, and Bogenberger (2023) evaluate two alternative strategies: at the intersection level and network level, for enforcing stop positions to reduce delays. Ye et al. (2020) develop a simulation-based genetic algorithm to identify parking lanes. Maxner et al. (2023) propose novel metrics for evaluating curbside performance and analyze the impact of different curb use allocations using simulation to aid decision-making. J. Liu, Ma, and Qian (2023) adopt a queuing model to reproduce the effects of congestion within an optimization problem for curbside management. Muriel et al. (2024) use SO with reinforcement learning to optimize route adjustments during deliveries to improve network efficiency. Burns, Michalek, and Samaras (2024) estimate the potential benefits of optimized curb management strategies in reducing double parking, congestion, and energy consumption through simulation-based analysis. These studies rely on traffic simulation (either at the algorithmic level or for validation purposes). Another solution approach is offered by Simoni and Claudel (2017) who model the interactions of freight vehicles by combining the Lighthill–Whitham–Richards (LWR) (Lighthill and Whitham 1955; Richards 1956) model with moving bottlenecks' theory, and integrating it into a meta-heuristic. Simoni (2024) further extends this approach into an agent-based framework to reproduce the effects of parking and cruising, and mitigate the congestion effects through parking guidance and geofencing strategies.

Some additional relevant studies concerning curbside management, with a focus on freight and ride-hailing, do not involve traffic models. McLeod and Cherrett (2011) investigate the concept of an advanced booking system for commercial bays based on a set of predefined rules. Yang, Roca-Riu, and Menéndez (2019) present an auction-based approach that maximizes the social welfare of the system by optimizing the utilization of pre-booked loading and unloading facilities. Mor, Speranza, and Viegas (2020) offer a broader viewpoint by incorporating the routing and scheduling component of the carriers into the management problem and its impact on the best assignment. Ranjbari et al. (2021) perform a regression analysis to quantify the benefits of a geo-fence from historical real-world data of one neighbourhood in Seattle. X. Liu et al. (2024) propose a re-routing strategy based on an inference approach for the pick-up and drop-off traffic congestion effects. Lim and Masoud (2024) addresses the dynamic usage allocation problem by modelling it as a Stackelberg leader-follower game and solving the corresponding bi-level optimization problem. Jain et al. (2024) develop a machine learning-based framework to predict curbside parking availability for commercial vehicles. J. Liu and Qian (2024) propose a framework for modelling multimodal curbside usage within a dynamic network, capturing route choices, curb user interaction, and spatiotemporal mobility patterns. Xu and Sun (2024) analyze urban curb parking dynamics under increased delivery demand using continuum models, evaluating the impacts of pricing, duration caps, and delivery bays on system equilibrium and management efficiency.

The related problem of commercial bays' location and sizing has also been investigated with different simulation and/or optimization approaches (Alho et al. 2018; Comi et al. 2017; Letnik et al. 2018; Pinto, Lagorio, and Golini 2019).

In addition to studies related to curbside management, some papers focus on the optimal control of vehicle trajectories, both speed and stoppages, for improving the state of traffic on roadways. These include studies that use connected and autonomous vehicles as moving bottlenecks to reduce traffic flow to help dissipate traffic jams. Proportional-integral feedback control (Piacentini, Pasquale, et al. 2019), nonlinear MPC (Piacentini, Ferrara, et al. 2019; Piacentini, Goatin, and Ferrara 2018), and Genetic Algorithm (GA) (Liard, Stern, and Delle Monache 2023) are among popular control mechanisms used for this purpose. These approaches utilize the first-order LWR model implemented through the Cell Transmission Model (Daganzo 1995) or a wave-front tracking approach (Liard, Stern, and Delle Monache 2023) which can become computationally intensive and time-consuming solution approaches in certain applications.

SO is a popular method for solving optimal control problems in several fields, including various areas of transportation such as urban rail transit planning (Hassannayebi et al. 2021), travel behaviour modelling (Xiong et al. 2018), and autonomous vehicles' testing (Wang et al. 2022; H. Zhang et al. 2022) among others. Iterative approaches involving multiple simulations (until convergence criteria are met) are usually time-consuming. As a result, surrogate-based optimization strategies that eliminate the requirement for several simulation runs have gained popularity and have been extensively investigated in the literature. Readers are referred to do Amaral et al. (2022) for a systematic review of various studies on surrogate modelling (or metamodelling)-based optimization. Surrogate-based optimization has also been employed in the transportation field. For example, Adacher (2012) solves the problem of traffic signal synchronization by transforming the original optimization problem into a surrogate form which can be solved efficiently using a gradient-based approach. X. Chen et al. (2014) compare several surrogate-based optimization strategies for optimizing highway toll charges to minimize a network average travel time. A surrogate-based optimization method using a regression Kriging model is proposed in X. M. Chen et al. (2016) to solve a congestion pricing problem, which requires intense computation of the corresponding objective function since it does not have a closed-form solution. A surrogate-based cooperative signal optimization method is proposed in Liang et al. (2021) using a radial basis function by assuming the sharing of information between different sub-networks. While most surrogate-based optimization studies utilize purely statistical models, there is a subcategory that combines analytical metamodels with a simple statistical model like a linear or polynomial regression model (Osorio 2019; Osorio and Bierlaire 2013; Osorio and Chong 2012; C. Zhang, Osorio, and Flötteröd 2017). The idea behind these surrogate models is to use subject knowledge associated with the simulation as a fast analytical model to help decision-making without completely relying on a statistical model. Further, a simple statistical model is used to learn the difference between the analytical model and the simulation. These studies have the benefit of not having to entirely rely on *black-box* statistical methods as well as using efficient optimization methods like gradient-descent (Lemaréchal 2012) given a closed-form differentiable surrogate model.

Although there have been advancements in the field of curbside management, some gaps remain that our research aims to address. Most simulation-based studies rely on optimization approaches that require extensive computational resources, limiting their applicability for real-time curbside management. This study extends the relatively new concept of surrogate-based optimization to the problem of curbside management to control vehicle stop locations and minimize the impact of congestion. Additionally, it is one of the few studies to systematically investigate the suitability of alternative surrogate models within the context of traffic optimization. While analytical metamodel-based optimization approaches are attractive for the reasons mentioned above, it is non-trivial to find a closed-form analytical model for every problem setup, which is the case with the setup presented in this article. Hence, in this work, we rely on a purely statistical approach. We select several available techniques from the literature and compare them based on accuracy and time consumption for solving a vehicle stop position control problem introduced in the following sections.

3. Problem definition and formulation

The control problem considered in this work can be defined as follows:

Given a set of vehicles destined to make stops on a segment of the road, control their longitudinal stop positions from the upstream end of the segment to minimize the adverse effects of the stops on traffic flow.

Throughout this work, the vehicles referred to in the above problem such as trucks, taxis, or ride-hailing cars approaching the road segment of their drop-off or pick-up operation are called *approaching vehicles* and the corresponding road segment is called the *stop segment*. It is assumed that each such vehicle is connected to an operator who identifies and transmits a stop position to the vehicles with the help of a controller using the knowledge of each vehicle's position and the associated traffic dynamics. The conceptual framework for the controller is illustrated in Figure 1. In the figure, the block 'Real World Traffic Evolution' captures the real-time state of the traffic in the system including

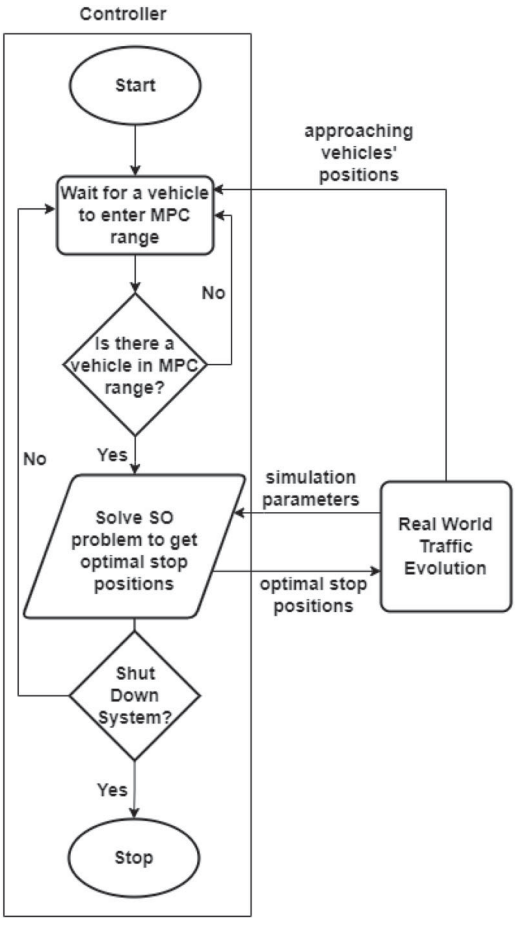


Figure 1. Flowchart of controller implementation.

the stop segment and its upstream and downstream conditions. The state of the system is assumed to be collected through sensors and communicated to the network operator. This information is used to set up the simulation in the SO optimization problem. Once solved, the resulting optimal stop positions are communicated to the respective vehicles in the traffic system who stop at the recommended locations thereby influencing the future evolution of traffic. The controller is formulated in the MPC framework wherein the optimal stop positions are decided based on their impact on a predefined objective function over a specified time period into the future, referred to as the *prediction horizon*. This horizon defines the time window within which future traffic conditions are considered during optimization. The problem is solved at fixed intervals (referred to as the *control horizon*, which defines the frequency at which the optimization problem is solved and updates are provided), and updated stop positions are communicated to the approaching vehicles based on the latest knowledge about the traffic conditions.

Recommendations on the stop positions are allowed to be updated until the vehicle enters the stop segment and no later to avoid confusing the drivers. The term *MPC range* refers to a time duration such that the solution to the SO problem is triggered only if there is an approaching vehicle expected to enter the stop segment within that duration. The vehicle's current distance to the stop segment and the current traffic speed are used to decide if the vehicle is within the MPC range. The goal is to ensure that the optimization is initiated in time to influence the vehicle's stop behaviour. In practice,

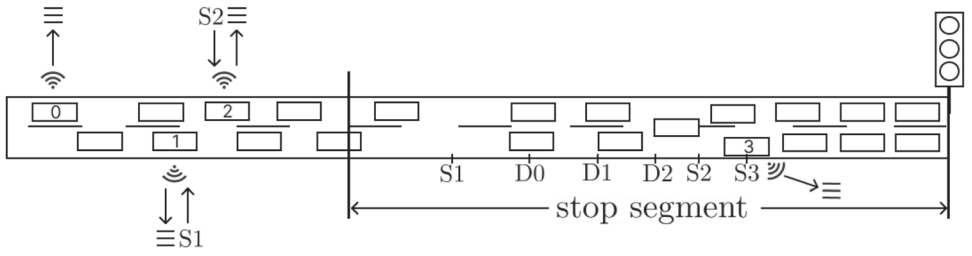


Figure 2. Schematic of a 2-lane stop segment with a traffic light at the downstream end.

the MPC range may be influenced by factors such as the time needed to solve the problem, the reaction time of the driver, the complexity of the approach, and communication delays. A schematic of the stop segment with approaching vehicles is presented in Figure 2. Vehicles 1 and 2 represent approaching vehicles within the MPC range, Vehicle 3 represents vehicles already stopped along the curbside forcing vehicles behind them to change lanes, and Vehicle 0 represents approaching vehicles outside the MPC range. The numbered vehicles relay their current state represented as the three stacked horizontal lines to the network operator. Notations D0, D1, and D2 represent the originally desired stop positions of Vehicles 0, 1, and 2, respectively. The network operator sends the optimal stop positions represented by S1 and S2 to Vehicles 1 and 2, respectively. Vehicle 0 is outside the MPC range and hence is not considered in the control problem and does not receive a suggested stop position.

In this study, we consider a macroscopic simulation framework defined by the LWR model which is used to model the flow of traffic on the stop segment along with a microscopically inspired framework for simulating the dynamics associated with vehicle stops resulting from traffic signals or pick-up/drop-off operations (Simoni and Claudel 2018). The framework is described briefly in the ensuing section followed by a detailed description of the SO problem and the proposed solution approach.

3.1. Traffic simulation framework

The selected simulation approach consists of a hybrid model where the main flows are modelled macroscopically, according to the LWR model whereas the dropoff/pick-up operations are microscopically modelled as temporary bottlenecks based on Simoni and Claudel (2018).

The LWR model is given as follows:

$$\frac{\partial \rho(t,x)}{\partial t} + \frac{\partial Q(\rho(t,x))}{\partial x} = 0 \tag{1}$$

where, for a given time t and position x, $\rho(t,x)$ represents the traffic density in vehicles per unit length and $Q(\rho(t,x))$ corresponds to the Fundamental diagram (FD) which maps the traffic density to traffic flow ($q(t,x)$) defined in vehicles per unit time. The FD is a function defined on $[0, \rho_m]$ where ρ_m is the maximal density (or jam density). In this study we adopt the triangular FD (Daganzo 1994) which is defined as follows:

$$q(t,x) = Q(\rho(t,x)) := \begin{cases} v_f \rho(t,x) & \text{if } \rho(t,x) \le \rho_c \\ w_c(\rho(t,x) - \rho_m) & \text{otherwise,} \end{cases} \tag{2}$$

where v_f denotes the free-flow speed of traffic, w_c denotes the congestion wave speed, and ρ_c denotes the critical density such that the maximum traffic flow or capacity $q_m = v_f \rho_c$. The spatial domain of the stop segment is defined as $[0, L]$ meters, and the time domain is defined as $[0, T]$ seconds.

In this paper, we adopt a Hamilton–Jacobi (HJ) partial differential equation (PDE) formulation of the LWR model and we solve it by adopting a Lax–Hopf formulation (Hopf 1969; Lax 1957). The HJ PDE is

expressed as follows:

$$\frac{\partial M(t,x)}{\partial t} + Q\left(-\frac{\partial M(t,x)}{\partial x}\right) = 0 \tag{3}$$

where $M(t,x)$ represents the cumulative vehicle count (Moskovitz function) for given time t and position x. To solve the HJ PDE (3) we need to define *value conditions* which are lower semi-continuous functions defined on a subset of $[0,T] \times [0,L]$ and can be represented by $\mathbf{c}(\cdot,\cdot)$. For each value condition $\mathbf{c}(\cdot,\cdot)$, a partial solution (Mazare et al. 2011) to the HJ PDE (3) can be defined using the Lax–Hopf formula (Aubin, Bayen, and Saint-Pierre 2008; Claudel and Bayen 2010a). Let $Q^*(\cdot)$ denote the Legendre–Fenchel transform of the FD (2) defined as follows:

$$Q^*(u) := \sup_{\rho \in [0,\rho_m]} [p \cdot u + Q(\rho)]. \tag{4}$$

Then the partial solution $\mathbf{M_c}(\cdot,\cdot)$ to (3) associated with $\mathbf{c}(\cdot,\cdot)$ can be represented (Aubin, Bayen, and Saint-Pierre 2008; Claudel and Bayen 2010a) by:

$$M_c(t,x) = \inf_{(u,\tau)\in \mathrm{Dom}(Q^*)\times\mathbb{R}_+} \left(c(t-\tau, x+\tau u) + \tau Q^*(u)\right). \tag{5}$$

The complete solution $\mathbf{M}(\cdot,\cdot)$ to the LWR model corresponding to a finite number of value conditions belonging to a set J can be defined using the inf-morphism property (Aubin, Bayen, and Saint-Pierre 2008; Claudel and Bayen 2010a, 2010b) as follows:

$$\forall (t,x) \in [0,T] \times [0,L], \quad M(t,x) = \min_{j\in J} M_{\mathbf{c}_j}(t,x) \tag{6}$$

Using the solution $M(t,x)$ across the entire simulation domain the density and flow can be constructed using $\rho(t,x) = -\frac{\partial M(t,x)}{\partial x}$ and $q(t,x) = \frac{\partial M(t,x)}{\partial t}$. The value conditions considered in this work over the domain $[0,T] \times [0,L]$ are defined as follows:

$$c(t,x) = \begin{cases} c^l_{ini}(x) & t = 0, \;\; x \in [(l-1)\Delta x, l\Delta x] \\ c^j_{up}(t) & x = 0, \;\; t \in [(j-1)\Delta t, j\Delta t] \\ c^k_{down}(t) & x = L, \;\; t \in [(j-1)\Delta t, j\Delta t] \\ c^n_{int}(t,x) & x \in [x_{b,n}, x_{e,n}], \;\; t \in [t_{b,n}, t_{e,n}]. \end{cases} \tag{7}$$

Here, $c^l_{ini}(\cdot)$ represents an initial density condition such that $l \in \{1,\ldots,\frac{L}{\Delta x}\}$ where Δx is the length of each initial density condition, $c^j_{up}(\cdot)$ and $c^j_{down}(\cdot)$ represent upstream and downstream boundary flow conditions such that $j \in \{1,\ldots,\frac{T}{\Delta t}\}$ where Δt represents the duration of each boundary flow condition, and $c^n_{int}(\cdot,\cdot)$ represents internal boundary conditions such that $n \in \{1,\ldots,n_i\}$ where n_i denotes the number of internal boundary conditions considered in the current simulation, $x_{b,n}$ and $t_{b,n}$ denote the position and time of the start of the internal condition and $x_{e,n}$ and $t_{e,n}$ denote the position and time of the end of the condition. Figure 3 presents a schematic of the different value condition blocks on a road segment.

The minimum value of the length L of the stop segment is decided by the Courant–Friedrichs–Lewy (CFL) condition (Courant, Friedrichs, and Lewy 1967): $\Delta t \leq L/v_f$. This ensures that the traffic waves originating from the upstream and downstream boundary conditions of the stop segment do not reach the opposite boundary within the duration of the flow conditions which would otherwise make the simulation unstable.

In the above, each initial density condition $c^l_{ini}(\cdot)$ is associated with a density $\rho_{ini}(l)$ which is assumed to be known prior to the simulation. Similarly, the boundary flow conditions $c^j_{up}(\cdot)$ and $c^j_{down}(\cdot)$ are associated with upstream and downstream flows $q_{in}(j)$ and $q_{out}(j)$, respectively. The traffic demand

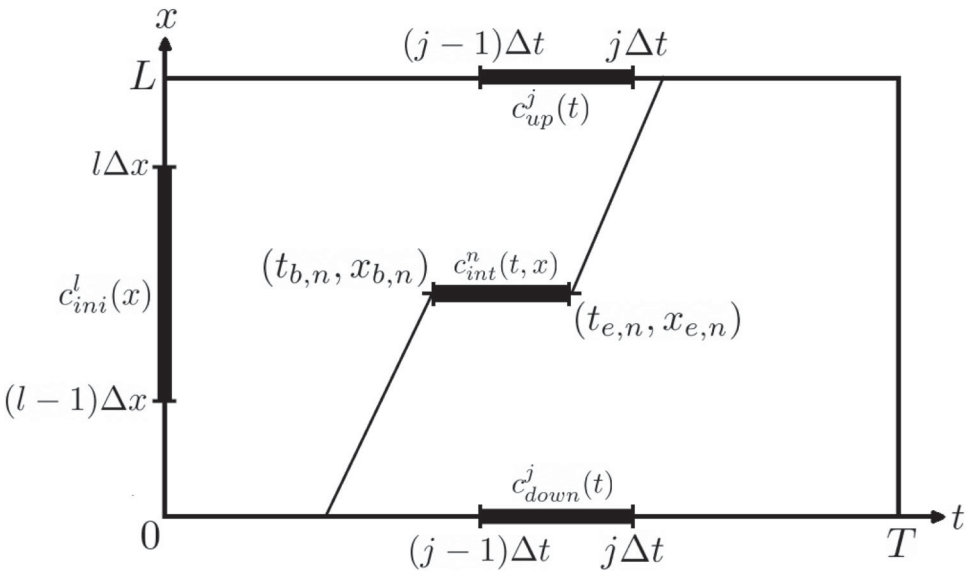


Figure 3. Schematic of different value condition blocks on a road segment.

$D(j)$ wanting to enter the segment over the span of the upstream boundary condition j, and downstream supply $S(j)$ allowed to exit from the segment during the span of the downstream boundary condition j are assumed to be known. The respective flows can then be calculated as follows:

$$q_{in}(j) = \min\left\{D(j), \frac{M(j\Delta t, 0) - M((j-1)\Delta t, 0)}{\Delta t}\right\}, \tag{8}$$

$$q_{out}(j) = \min\left\{S(j), \frac{M(j\Delta t, L) - M((j-1)\Delta t, L)}{\Delta t}\right\}. \tag{9}$$

Each internal condition $c^n_{int}(\cdot,\cdot)$ is associated with a starting and ending value of the cumulative number of vehicles. In this work, the internal conditions correspond to the bottlenecks created by vehicles stopping for pick-up or drop-off operations. The stopping vehicle acts as a lane blocker forcing vehicles to use the remaining lanes to pass, resulting in a drop in the maximum flow through that cross section of the road. This phenomenon is captured by incorporating internal flow conditions which govern the flow through a section of the road. Note that an internal condition block is only applied when the through traffic volume is greater than the flow that can be accommodated by the remaining lanes. The trajectories for vehicles intending to carry out such operations are tracked from the time of their entry into the segment up to when the operation is completed. The computation of the spatiotemporal domain and the cumulative number of vehicles at the extremes of the internal conditions is carried out using Algorithms 1 and 2 from Simoni and Claudel (2017) given prior knowledge of the time of entry into the stop segment, desired stop position, and stop duration. The mathematical expressions relating the various types of conditions to aforementioned quantities as well as the approach to solve (5) for the different types of conditions can also be found in Simoni and Claudel (2017). Figure 4 presents a simulated example of how vehicle stoppages are considered to impact the traffic flow in this study. Here, $v_f = 14$ m/s, $w_c = 2.8$ m/s, $\rho_c = 0.04$ veh/m, and $\rho_m = 0.24$ veh/m. Notice, that when there is no stopped vehicle, the density is equal to ρ_c. When the vehicle stops, the density downstream of the stop becomes half of the critical density indicating a total flow in that region to be half of capacity due to the 1 blocked lane. The upstream density here can be computed from the triangular diagram by solving $(\rho_m - x)w_c = q_m/2$ for x which comes out to 0.14 veh/m, as observed in the figure. Note that

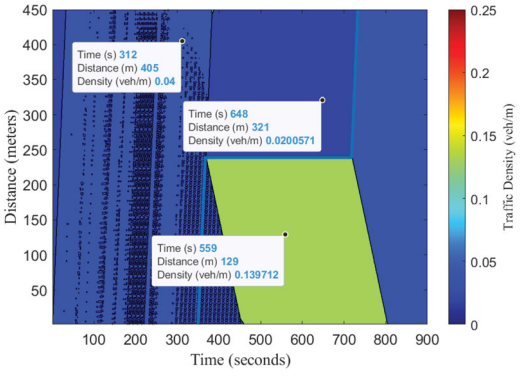


Figure 4. Simulation example to depict the impact of vehicle stoppage on traffic density.

Table 1. Notation used for SO problem formulation.

Parameter/Variable	Description
t	Time at which control is executed
N_p	Prediction horizon duration (time steps into the future up to which the impact of control is evaluated)
$N(t)$	Number of approaching vehicles at time t
x_i	Longitudinal stop position for vehicle i
X_t	Vector of stop positions for approaching vehicles such that $X_t = \{x_1, \ldots, x_{N(t)}\}$
u_t	Non-controlled simulation parameters for the prediction period starting at time t
k	Time step index within prediction horizon
$f(X_t, u_t)$	Objective function computed from simulation
$q_{\text{out}}(k; X_t, u_t)$	Simulated traffic outflow over time-step k
$q_{\text{in}}(k; X_t, u_t)$	Simulated traffic inflow over time-step k
$D(k)$	Upstream traffic demand for time-step k
X_D	Vector of desired stop positions for approaching vehicles
X_{US}, X_{DS}	Vectors of maximum distance upstream and downstream, respectively from X_D based on operation requirements
w_{SB}	Objective penalty weight on congestion spillback
W_D	Vector of objective penalty weights corresponding to detour from desired stop positions

the slight deviation in the upstream and downstream density values from their theoretical values is due to rounding errors in computations and does not significantly impact the results.

Overall, in the above framework, the input parameters required to perform a simulation over a given prediction horizon consist of the initial traffic conditions, the upstream and downstream traffic conditions, the timing of the traffic signals (positions of signals are fixed), and the anticipated arrival time, desired stop position, and stop durations of the approaching vehicles. Vehicles already present on the stop segment that perform stops at various positions on the segment are considered as well. The next section presents the SO problem description considering the above simulation framework.

3.2. Simulation-based optimization formulation

The controller is formulated in the MPC framework where an optimal decision is made by predicting the future behaviour of the system over a predefined horizon with the help of a model while considering the control objectives and constraints. Here, the network operator aims to regulate the stop positions of approaching vehicles which form the decision vector to minimize their adverse effects on the traffic flow. The notation utilized for defining the optimization problem is presented in Table 1.

The SO problem can be defined as follows:

$$\begin{aligned}
\min_{X_t} \quad & f(X_t, u_t) \\
& = -\sum_{k=1}^{N_p} q_{\text{out}}(k; X_t, u_t) \\
& + w_{SB} \sum_{k=1}^{N_p} (D(k) - q_{\text{in}}(k; X_t, u_t)) \\
& + W_D \cdot |X_D - X_t| \qquad (10) \\
\text{such that} \quad & x_i \in [w_c \Delta t, L - v_f \Delta t] \ \ \forall i \in \{1, \ldots, N(t)\} \qquad (11) \\
& X_t \in [X_D - X_{US}, X_D + X_{DS}] \qquad (12)
\end{aligned}$$

Here, (10) is the objective function which is to be minimized. The first component represents the total traffic outflow over the complete prediction horizon, the second component reflects the traffic demand that is blocked by the stop segment's traffic, as a result of congestion spillback upstream, and the third component accounts for the cost of detour from the initially desired stop positions for all approaching vehicles. The magnitude of the detour penalty for each approaching vehicle can be set differently by changing corresponding elements in the vector W_D. Here, the first two terms directly aim to reduce the impact of stoppages on the traffic flow. The third term covers the cost of inconvenience caused by the detour in stopping at a different position than desired for a given operation. For instance, such costs could include a reward that the ride-hailing service provider may need to pay its customers for inconvenience resulting from changes to their pick-up and drop-off positions in order to show their commitment to customer satisfaction. The service provider would need to be compensated for such a reward by the government authorities desiring improvement in traffic flow.

Equation (11) specifies the bounds on the decision vector required to maintain the stability of the model. This considers the impact of the internal conditions on boundary conditions preventing their interaction within the time step in which they originate. The minimum distance of the stop position from the upstream boundary is given by the distance travelled by a congestion wave over one simulation time step. Similarly, the minimum distance of the stop position from the downstream end is given by the distance travelled by a free-flowing traffic wave over a single time step. Equation (12) prevents unreasonably large changes in stop positions from the initially desired positions. These constraints are useful when the improvement in traffic flow achieved by large changes in stop positions is significantly higher than the detour penalty which would result in the optimal stop positions being far from the desired position. Regardless of the improvement, certain operations cannot be performed more than a certain distance away from the desired position which is accounted for through these bounds. Depending on the time resolution for simulation, the candidate stop positions can be further reduced to a set of discrete values to reduce the search space for the solution. For instance, with a time resolution of 1 s, the stop positions are approximately in multiples of v_f.

The objective function (10) is computed from the simulation described in Section 3.1 for different X_t given a set of non-controlled simulation parameters u_t. Here, u_t represents all possible parameters that affect the simulation including spatial limits of the segment, FD parameters, initial density, upstream demand $D(k)$, downstream supply, traffic signal cycle times, approaching vehicles' arrival time, and approaching vehicles' stop durations. The simulation also considers vehicles intending to make a stop and already present on the stop segment at the beginning of the simulation although their stop position is not included in the decision variables. Hereafter, these are referred to as *on-segment vehicles*. The stop positions and corresponding durations of the on-segment vehicles are considered part of u_t.

The objective function (10) lacks a direct analytical gradient due to the use of simulation for evaluation. Therefore, the SO problem (10)–(12) cannot be solved using efficient approaches like gradient

descent. Instead, meta-heuristic evolutionary algorithms such as GA can be used to find an optimal solution where the values of X_t are iteratively improved in order to reach well-performing solutions. When adopting meta-heuristics, the objective function iteratively updates until a stopping criterion is met. Since each evaluation of the objective function (10) requires a simulation, meta-heuristics would likely require significant computation efforts to achieve high-quality solutions. In this study, we replace (10) with an approximate function known as a surrogate model which can ideally be evaluated faster than the actual simulation resulting in reduced computational requirements when implemented within a GA. This forms the basis of the surrogate-based solution approach for the SO problem which is discussed in the following section.

4. Surrogate modelling

In the proposed approach, the objective function (10) is estimated through a suitable surrogate model that produces an approximation of (10). The goal is to remove the need for simulation in evaluating the objective function when solving the SO problem (10)–(12) with an evolutionary algorithm like GA. This study considers two types of surrogate models- *local* models and *global* models. *Local* models are created for individual runs of the controller, are trained online, and require a new training sample each time the SO problem is solved. *Global* models on the other hand are created only once and are trained offline using a large training sample that is created in advance considering plausible scenarios that would arise during different individual controller runs. The first phase of model development is to determine the model's inputs and outputs. Section 4.1 describes the inputs considered by both types of models and the corresponding sampling technique for training and testing. Additionally, several aspects of the training procedure are highlighted. Finally, the performance of the surrogate models is evaluated by means of alternative metrics in Section 4.3.

4.1. Input and sampling

To solve the SO problem using an evolutionary algorithm, like a GA, multiple feasible solutions are ranked by evaluating them through the objective function. The surrogate model output is intended to replace the objective function when solving the SO problem with a GA. The model takes the set of decision variables X_t as input and generates an output equivalent to the corresponding value of the objective function. In the MPC framework, the SO problem is solved whenever an approaching vehicle is detected within the MPC range. The number of approaching vehicles at any given time affects the length of X_t and consequently the number of inputs for the surrogate model. The objective function (10) also includes the non-controlled simulation parameters u_t as inputs that should be considered in the surrogate model. For *local* models, the effect of a given u_t on the surrogate model output is captured implicitly in the simulations used to produce the dependent variables in the training data. For *global* models, since u_t varies across different controller runs, the parameters included in u_t mentioned in Section 3.2 are also taken as inputs to these models. To account for the changing number of decision variables, multiple *global* surrogate models with different numbers of inputs can be trained offline and can be applied to cases with different numbers of approaching vehicles as required.

For *local* models, each data point in the training sample consists of a distinct set of stop positions X_t and the corresponding output from (10) given a fixed u_t. For each data point, every stop position is selected randomly and independently from a set of predefined positions which are based on the minimum distance travelled by a vehicle in free flow over one time step as the vehicles stop very close to these positions in simulation. For instance, for free-flow speed v_f and time step 1 s, the stop positions are selected from the set $\{v_f, 2v_f, 3v_f, \ldots\}$ while obeying the stability constraints defined in Section 3.2.

Global surrogate models, should ideally be able to estimate well the output from (10) for any u_t and X_t that may occur across different runs of the SO problem during MPC. For each data point, the stop positions are selected as discussed above. The sampling for parameters within u_t is discussed in the remainder of this section. For the stop duration, it is reasonable to select possible values from

multiples of 30 s(this type of information would be realistically available from the driver). The stop durations are randomly selected from multiples of 30 s up to a reasonable upper bound. For all other parameters, the values are selected randomly from uniform distributions with the following bounds- $[0, q_m]$ for the demand and supply parameters, $[0, \rho_m]$ for the initial density parameters, and $[-T_r, T_g]$ for the red signal start time where T_r and T_g are the magnitude of the red time and green time in a traffic signal cycle respectively. The negative lower bound allows for cases in which the start time of the prediction horizon is in the middle of the red phase of the signal. The start and stop positions, and stop durations of the on-segment vehicles are selected in the same way as those of the approaching vehicles. Naturally, the start position values are smaller than the stop position values.

4.2. Surrogate model selection

The following methods are considered for surrogate modelling:

(1) Linear regression (LR)
(2) Polynomial regression (PR)
(3) Regression-based tree ensemble (RTE)
(4) Shallow neural network (NN)
(5) Radial basis neural network (RBNN)
(6) Gaussian process regression (GPR)

A short description of the different methods investigated in this study is provided in Appendix 1.

Among the above techniques, LR and PR are selected as surrogate models as they result in simple linear and quadratic optimization problems when used as surrogates (linear and quadratic mixed integer problems when the decision variables are discrete). These can be solved using freely available standard solvers, making them highly suitable for real-time control. RTE, NN with a sigmoid *transfer function*, and RBNN with the radial basis function, are considered valuable for capturing complex nonlinear relationships between inputs and output. GPR is selected due to its popularity in surrogate-based optimization studies, as evident in X. Chen et al. (2014), as well as its demonstrated ability to outperform NNs (Tavassoli, Waghei, and Nazemi 2022). However, it is important to note that these techniques require the use of meta-heuristics or heuristic algorithms to solve the optimization problem. Since straightforward methods like LR and PR are often unable to effectively learn complex relationships between inputs and output like the other candidate techniques, there exists a trade-off between the accuracy and time required to find a well-performing solution. In the analysis section, we further assess the models in terms of their usefulness in real-time control.

4.3. Analysis of surrogate models

4.3.1. Comparison metrics

When comparing various surrogate modelling techniques, it is essential to use an evaluation metric that reflects the purpose of the model. In this study, the goal of the surrogate model is to replace the simulation in the solution to the SO problem. The SO problem is solved by means of a meta-heuristic algorithm that relies on accurately ranking the problem solutions based on the corresponding values of the objective function. Therefore, it is important that the outputs of the surrogate model can be used to accurately rank the solutions even if the magnitude of the outputs themselves is not close to the outputs of the simulation. Here, correct ranking implies that solutions are correctly ordered by the magnitude of the objective value produced in the simulation. As a result, a standard metric such as the Root Mean Squared Error (RMSE), which evaluates the model based only on the magnitude of the outputs, would not be sufficient for the purpose of this study. A suitable metric for our problem is the Ranking Error (RE) (Joachims 2005), which evaluates the surrogate model based on the number of incorrectly ordered pairs of solutions when the solutions are ordered according to the corresponding

Table 2. Combination of the number of vehicles considered in different cases for testing.

Combination name	C1	C2	C3	C4	C5
Approaching vehicles	4	4	4	6	6
On-segment vehicles	0	2	4	0	2

model outputs. It is mathematically defined as follows:

$$\mathrm{RE} = \frac{n_{SP}}{N(t)^2}, \tag{13}$$

where n_{SP} denotes the number of *swapped pairs* (pairs of solutions which are incorrectly ordered) obtained from:

$$n_{SP} = |\{(i,j) : (f_t(X_t^i) - f_t(X_t^j)) \times (\hat{f}_t(X_t^i) - \hat{f}_t(X_t^j)) < 0\}| \tag{14}$$

where $\hat{f}_t(\cdot)$ denotes the output of the surrogate model, and $N(t)$ is the number of solutions in the validation set. A larger RE implies that a larger number of solutions is incorrectly ranked therefore resulting in a higher possibility of the meta-heuristic producing a sub-optimal solution.

Since the goal of solving the SO problem is real-time traffic control, it is important that the optimization solution can be derived within a reasonable time as per the requirements of control (e.g. a few seconds). In the case of *local* models, the total time required to reach an optimal solution includes the time to produce training samples, train the model, and solve the problem using the trained model. A model that results in close to optimal solutions but requires a large number of samples to train to achieve that level of accuracy, or if the meta-heuristic takes a long time to converge using the model, may not be particularly useful for real-time control. Therefore, besides the RE, we also consider the solution time as an evaluation metric for the model choice. In the case of *global* models, since the model is trained offline, this time excludes the time to produce the samples and train the model.

4.3.2. Analysis of local models

To test the performance of the surrogate models under different scenarios, we create several test cases characterized by different numbers of approaching vehicles, different numbers of on-segment vehicles, and different values of all considered simulation parameters. Since the number of approaching and on-segment vehicles directly affects the number of simulation parameters, scenarios are first broadly classified by the number of both types of vehicles, and following that sub-scenarios are defined based on different values of simulation parameters. Scenarios based on the combinations of the numbers of vehicles are denoted by C1, C2, C3, C4, and C5, which are briefly described in Table 2. Further, 12 sub-scenarios are considered for scenarios C1 and C4 and 13 sub-scenarios for scenarios C2, C3, and C5. Details of the simulation settings for these sub-scenarios are presented in Appendix 2.

To determine the modelling methods that work better with less number of samples, we compare the performance of the models trained with different methods using sample sets of sizes 100, 200, 300, 400, and 500 produced as described in Section 4.1. For each number of samples, we produce 5 sample sets, and models are trained on each set to account for randomness in data generation. Table 3 presents the number of parameters trained in the case of each of the selected modelling methods. It is observed that GPR has the least number of parameters while RTE has the highest number of parameters. The number of parameters in the LR and PR models changes with the number of variables in the model. To compare the models trained using different methods, the models are applied to test sample sets with 1000 samples produced independently from the training samples but using the same sampling method. A test sample set is produced for each of the sub-scenarios within each scenario. An average RE is computed for each combination of modelling method, training sample size, and scenario (C1 to C5) by averaging the RE over the models produced from the 5 training sample sets for each sample size and sub-scenario.

Table 3. Average training time, time spent in solving the optimization problem, and the number of parameters trained for the different modelling techniques for *local* models.

Time (seconds)	LR	NN	RBNN	RTE	PR	GPR
Training	0.006	0.29	0.046	0.329	0.007	0.122
Solving	0.01	12.50	15.90	42.40	0.60	0.45
# Parameters	n+1	25	21	100c	$(n+1)(\frac{n}{2}+1)$	3

Here, n denotes the number of decision variables (4 or 6 depending on the scenario). c denotes the number of *cuts* that vary from tens to hundreds with an increasing number of training samples.

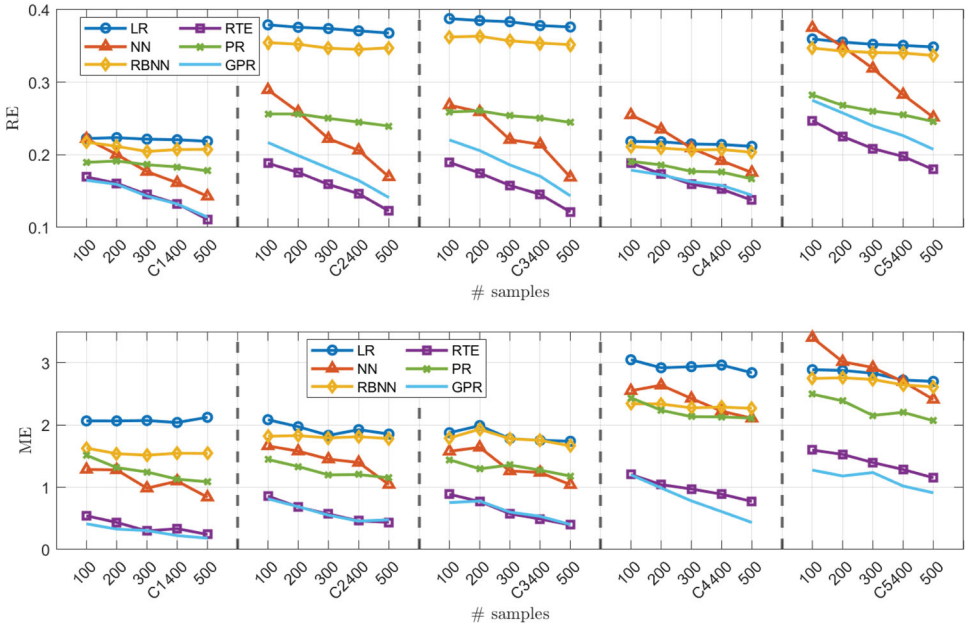


Figure 5. Average RE [top] and average ME [bottom] for the different modelling techniques and scenarios (C1–C5) for *local* models with a varying number of training samples.

The accuracy of the alternative models across the scenarios is illustrated in Figure 5. The x-axis represents the number of training samples used while the y-axis represents the RE. Each curve represents a modelling technique. Each point presents the average value over 5 models trained for the same training sample size and over the sub-scenarios within the scenario mentioned below the sample sizes on the x-axis. As expected, the performance of all the modelling techniques improves with increasing sizes of training samples. Particularly, the largest increase in accuracy is observed for NN which shows an average improvement of 0.10 in increasing the training samples from 100 to 500 across all scenarios, followed by RTE and GPR which both show an average improvement of 0.06. Among the different modelling techniques, RTE has the lowest RE for all scenarios followed by GPR. The least accurate model is LR followed by RBNN which hardly shows any improvement with the number of samples. Generally, the performance is worse for scenarios C2, C3, and C5, characterized by on-segment vehicles which is possible because of a more complex relationship between the decision variables and the parameters induced by the presence of these vehicles.

The difference between the objective function value obtained by solving the SO problem directly using the objective function (10) and that obtained by solving it using the surrogate models is also evaluated. The purpose is to validate the observations obtained from the RE since RE is only a proxy for the true performance in optimization. Again, a single error value is obtained for each combination

of modelling method, training set size, and scenario C1 to C5 by averaging the results from the 5 alternate models for each size and all sub-scenarios, for each modelling method. This is referred to as the Mean Error (ME) and is also presented in Figure 5 for the *local* models. The time step duration used for evaluating (10) is 10 s. In this study, the decision variables are restricted to take values in the multiples of the free-flow speed as described in Section 4.1. A solution using the LR model can be obtained simply by observing the signs of the decision variable coefficients. An integer programming solver is used to solve the optimization problem using the PR model. A GA is adopted for obtaining a solution to the SO problem using all other modelling methods. For the GA, the functional tolerance value is set to 0.5 which means that the algorithm terminates if the average change in the value of the objective function is below this value over a pre-defined number of generations which here is set to 50 by default. The other parameters are also kept as the default values for the inbuilt MATLAB function (The Mathworks Inc 2021a). The maximum run time for the GA is set at 30 min. Note that 30 min is already high for solving a real-time control problem, but it is adopted here to allow for the GA to terminate naturally by the specified termination criteria. In cases where the simulation-based objective is larger than the surrogate-based objective (due to GA not reaching the optimal solution using simulation), the error is taken as 0 since this kind of error is favorable in the current setting. It is observed in Figure 5 that between modelling techniques, the ME is lowest for RTE and GPR models and worst for RBNN and LR. This is consistent with the observation from examining the RE and shows that RE works as a reasonable proxy for actual error values in optimization. In general, the ME decreases with an increase in sample size although in some cases there is no significant trend such as LR and RBNN in scenario C1.

While accuracy measures such as the RE and ME provide an indication of the models' effectiveness in identifying high-performance solutions, the real-time control problem presents additional challenges. It is essential to evaluate the amount of time required to create the requisite number of training samples, train the surrogate model, and solve the optimization problem (by means of meta-heuristics or exact methods for LR and PR). In general, the time taken to train the models is negligible compared to the time for producing the samples. The mean training times for the different modelling techniques are presented in Table 3. The time spent in solving the optimization problem is also presented in Table 3. For solving the problem, LR and PR models are quite efficient as they rely on standard solvers. The RTE model takes the longest to converge using a meta-heuristic algorithm. This can be attributed to the tree structure which is more computationally intensive to evaluate for each candidate solution. Among the techniques that are integrated into meta-heuristic approaches, GPR is the fastest to converge with its solving time being comparable to that using the PR model. This is potentially because of the more straightforward evaluation of candidate solutions through GPR. Besides the solving time, the local models also need the time to produce the samples that are needed to train the models at the time of control. It is observed that the time to produce samples increases with the number of stopping vehicles such that 100 samples of scenarios C1 to C5 take around 9.5, 13.4, 17.5, 15.4, and 19.8 s, respectively when produced sequentially. To get a sense of time savings using the surrogate model, we need to compare the total time including sample generation, training, and solving to the time required to converge using the simulation directly. The average time taken by the simulation for scenarios C1 to C5 is 262, 312, 399, 552, and 601 s, respectively. Note that surrogate-based optimization still saves a lot of time as compared to simulation-based solution even with the online generation of training samples.

4.3.3. Analysis of global models and comparison with local models

Since *global* models are trained with a larger number of variables (including both the decision variables and other simulation parameters), we require a larger training sample to understand the influence of all variables on the objective function. Unlike *local* models, *global* models are trained to account for the variations in the simulation parameter values. Therefore the same model can be used in a broader range of test cases described in Appendix 2. Regardless, scenarios with different numbers of approaching and on-segment vehicles will still need separate models as they have different numbers of variables.

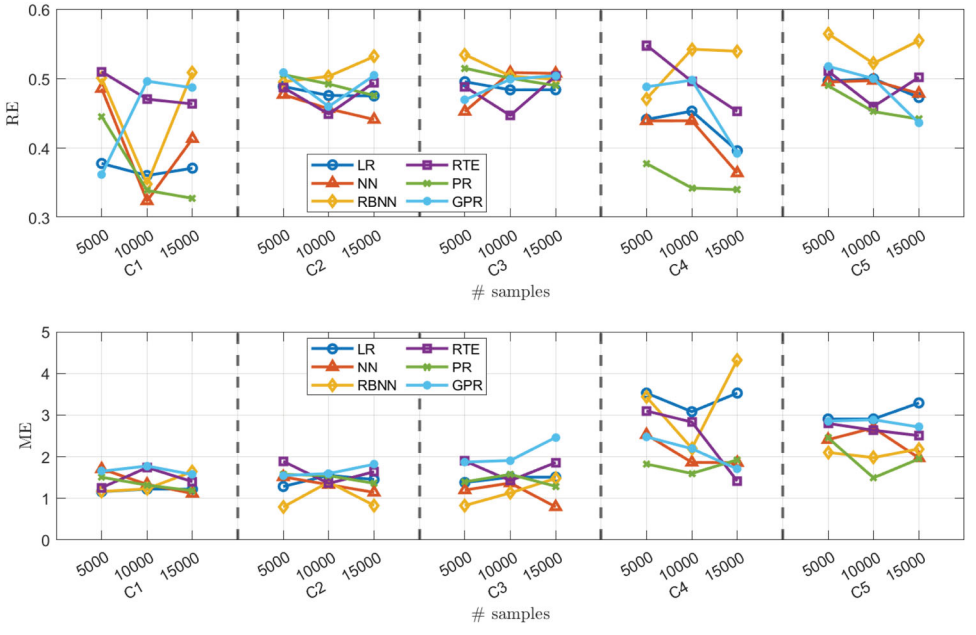


Figure 6. Average RE [top] and average ME [bottom] for the different modelling techniques and scenarios (C1–C5) for *global* models with a varying number of training samples.

In this case, we compare the performance of models trained using training sets of sizes 5000, 10,000, and 15,000 samples produced using the approach mentioned in Section 4.1. Testing is done using the same test sets described in the previous section for *local* models which are produced for individual sub-scenarios. The trends in the RE and ME for the *global* models with increasing training set sizes is presented in Figure 6 which is similar to Figure 5. In general, it is expected that an increase in the number of samples should result in an improvement in the performance of the model. However, in this case, as presented in Figure 6 there is no particular trend in RE with an increasing number of samples. A reason for this is that we only use a subset of the variables used to train the model in the testing process. Therefore, it is not guaranteed that the model necessarily learns better the impact of the subset of variables that we are interested in. In particular, models trained for scenarios C1 and C4 with no *on-segment vehicles* show smaller values for RE as compared to other scenarios. This is similar to the observation made for *local* models which also perform relatively better in the absence of *on-segment vehicles*. Overall, compared to Figure 5, the RE is much larger when using the *global* models as compared to the *local* models irrespective of the number of samples especially when compared for RTE and GPR models. Similarly, the ME for the *global* models presented in Figure 6 is also worse than the best results obtained using RTE and GPR models in the *local* case. The results with the other modelling techniques are similar to those obtained using the *local* models.

Since these models can be trained offline and stored for use whenever the corresponding scenario arises in the real world, the time taken to generate the required samples and further training these models is of no consequence to their real-time implementation. The only time duration that matters, in this case, is the solving time. The same is presented in Table 4. Interestingly, the meta-heuristic dependent models require more time than when used in a *local* model context. RTE method takes almost thrice the time taken using *local* models. This is possible because of the more complicated and hence more computationally intensive model obtained by training with a much larger set of variables which also includes the simulation parameters in this case. LR and PR still have comparable solving times to the *local* models as they are solved directly using a solver and do not depend on evaluations of the surrogate within a meta-heuristic. GPR still takes the lowest time among meta-heuristic dependent

Table 4. Average time spent in solving the optimization problem and the number of parameters trained for the different modelling techniques for *global* models.

Time (seconds)	LR	NN	RBNN	RTE	PR	GPR
Solving	0.013	24.10	19.49	124.3	0.46	10.09
# Parameters	n+1	577	841	100c	$(n+1)(\frac{n}{2}+1)$	3

Here, n denotes the sum of the number of variables in the model (including both decision variables and simulation parameters). n varies from 24 to 40 across scenarios. c denotes the number of cuts that vary from a few hundred to a few thousand with an increasing number of training samples.

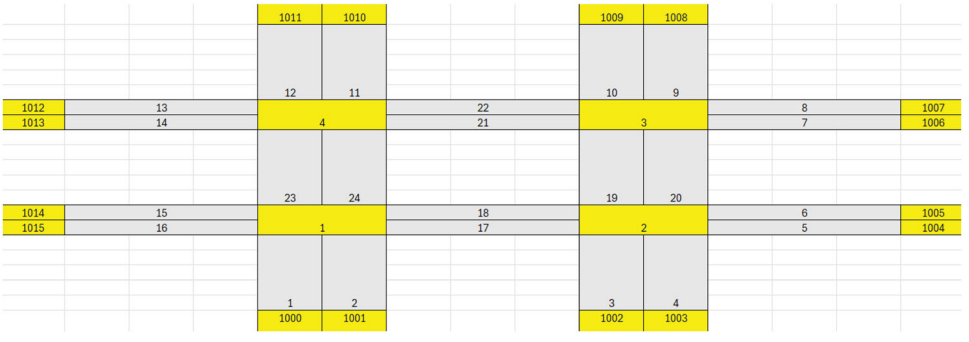


Figure 7. Network example with multiple stop segments.

methods although it is 20 times slower than when used in the *local* context. The number of parameters in the models is also presented in Table 4. The number of parameters in LR and PR models, in this case, is naturally higher than in the *local* context because of the higher number of variables in this case, while the number of parameters in the NN and RBNN models is higher because of the higher number of neurons used in this case to account for the larger number of variables. The number of parameters in GPR is the same as in the *local* case due to the use of the same basis and kernel functions (see Appendix 1 for details) as in the *local* case.

Overall, as expected, optimization using surrogate models has considerable computational advantages over simulation-based solution. The *local* models perform considerably better than the *global* models in terms of accuracy although they do require additional time to generate the samples and train the model. However, since the training samples are independent and do not need to be produced sequentially, it is possible to parallelize the simulations used to create the training samples thereby significantly reducing the overall required time. Therefore, real-time control should still be possible using these *local* models even though they require online training. In general, RTE and GPR perform the best in terms of accuracy, and between the two, GPR is also considerably faster. While LR and PR models are also quite fast, they are not accurate and hence are not recommended for solving the SO problem.

4.3.4. Analysis on a network example with multiple stop segments

The above sections evaluate the performance of various surrogate models for optimizing stop positions within a single stop segment. However, the method is flexible and not constrained by network geometry, allowing for its extension to optimization at a network level. To evaluate the scalability of the proposed approach, we present a small network example with multiple stop segments. The network structure is presented in Figure 7, where the yellow blocks represent the input and output nodes and intersections, while the gray blocks denote the network segments, each with two lanes. All road segments are 100 m in length, and the simulation duration is 600 s. Traffic parameters remain consistent with those in previous sections and are detailed in Appendix 2 along with a time step value of 2 s. The GA settings are also the same as used in the previous sections with the solver time limit set to 10 min.

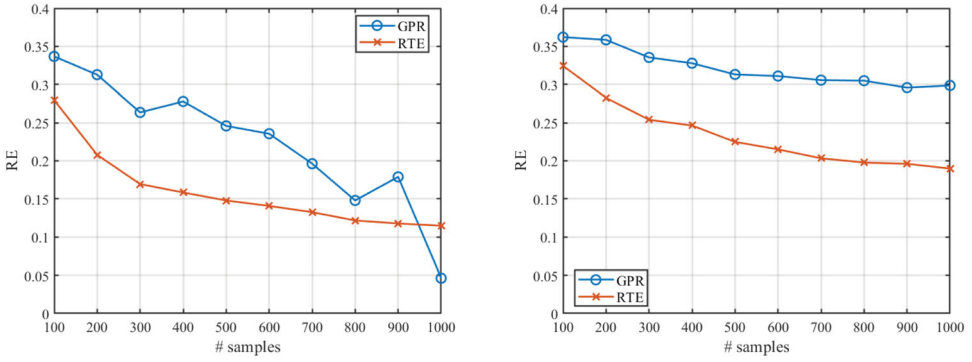


Figure 8. Average RE for GPR and RTE methods with [left] 2 approaching vehicles, and [right] 4 approaching vehicles, in the network example.

The incoming demand through all entry nodes is fixed at 0.5 veh/sec, and the supply is assumed to be unrestricted. Predefined turning proportions, assumed to be derived from historical data in real-world applications, are used at the four intersections. Initial conditions are also predefined. Traffic signals have a fixed cycle duration of 60 s, with a green phase of 36 s. The simulation starts with green lights at all intersections. Stopping vehicles are restricted to segments 18, 19, 21, and 24, corresponding to the central block of the network. This constraint reflects a choice to focus on delivery and pick-up/drop-off operations along the central block and does not imply a limitation of the proposed method. Trucks enter through prespecified links and make deliveries based on recommendations from the controller.

Two scenarios are considered, with varying numbers of trucks entering the network: NC1 with 2 vehicles, and NC2 with 4 vehicles. These scenarios aim to examine the general scalability of the proposed method over multiple stop segments and nodes. In this example, on-segment vehicles at the start of the simulation are not included, and no sub-scenarios with traffic parameter variations are considered. For this analysis, only local models are considered trained using GPR and RTE. These methods are selected based on their superior performance, as discussed in previous sections. For each scenario, both models are trained on datasets containing 100 to 1000 samples with 100 sample increments, with 5 random sets at each sample size. Each data point represents a unique solution with the prescribed stop segment and the stop position within that segment. A common test set of 1000 samples is used for evaluation in each scenario. Note that entry times, entry segments (as numbered in Figure 7), and stop duration for vehicles approaching the network are fixed across all points, as they are not decision variables. For the two vehicles in scenario NC1, these values are 30 s, 2, and 60 s, and 60 s, 14, and 60 s. For the two additional vehicles in scenario NC2, these values are 100 s, 10, and 60 s, and 40 s, 6, and 60 s, respectively. In this example, the spillback penalty is set to 10, and the detour penalty is set to 0.

Figure 8 presents the variation in RE for GPR and RTE for Scenarios NC1 and NC2. As observed in previous sections, the performance of the models deteriorates with an increasing number of decision variables which increases with the number of approaching vehicles. The performance is improved by considering more training samples for the surrogate models. Figures 9 and 10 present the variation in ME for GPR and RTE for Scenarios NC1 and NC2, respectively. The error in the absolute value of the objective is higher in this case compared to previous sections which is reasonable as the magnitude of the objective and the error can multiply with the number of segments in the network compared to a single segment in the previous sections. We also additionally present the mean percentage error for the models in both scenarios for better understanding of the magnitude of the error. The overall error in the objective using the surrogate-based approach is less than 2% for any number of training samples in both scenarios which is promising.

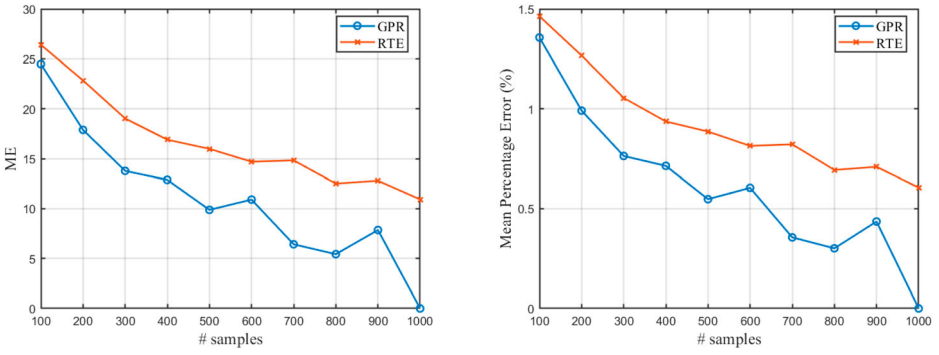


Figure 9. [left] Average ME, and [right] average Mean Percentage Error for GPR and RTE methods with 2 approaching vehicles in the network example.

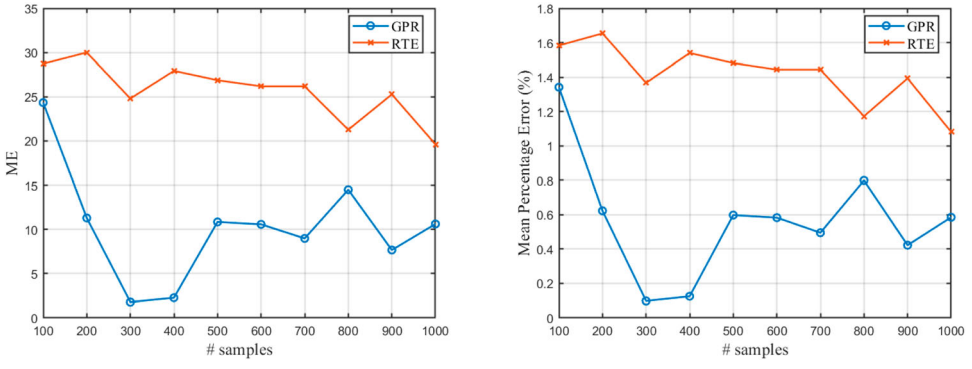


Figure 10. [left] Average ME, and [right] average Mean Percentage Error for GPR and RTE methods with 4 approaching vehicles in the network example.

The average time to solve the optimization problem using the GPR model is 2.61 s and RTE is 77.5 s across training samples and scenarios. The average time to train the GPR and RTE models is 0.81 s and 0.47 s, respectively. The average time to compute 100 simulation samples for NC1 and NC2 sequentially is 93.48 s and 97.05 s, respectively. Direct SO with GA hits the solver time limit of 600 s for both scenarios indicating that both the surrogate modelling approaches are an improvement over the SO solution in terms of convergence time. Note that simulation-based sample generation for surrogate model training is a parallelizable process unlike using the simulation directly as the fitness function within the GA. This would make it possible to use the proposed approach on network optimization instances with multiple segments and intersections by producing a larger number of training samples in parallel. This makes this method more feasible for real-time control compared to direct SO using GA. Another approach to making the solution scheme feasible for real-time control at a network level is decentralization involving dividing the network into zones and solving the problem for each zone independently with some information exchange between the zones. This can be a subject of future research.

In the following section, we present an example of a realistic application of the control implemented using the surrogate-based optimization approach using the GPR model versus directly using the simulation to solve the optimization problem.

5. Application: management of ride-hailing vehicles' pickup/drop-off operations

To illustrate the effectiveness of the proposed control approach on traffic system performance as well as to compare the surrogate-based approach for obtaining the optimal controls with the direct approach using the simulation, we present an example application for controlling the pick-up/drop-off positions of ride-hailing vehicles on a stretch of a signalized urban road. We also study the impact of using different magnitudes of the penalty weight for the detour caused by control. The *local* GPR surrogate model is selected for the analysis in this section due to its suitability to the optimization problem that emerged in Section 4.3.2. We test two training sample set sizes for the models, namely 50 and 500 samples. To demonstrate the effectiveness of the surrogate-based optimization with low computational cost, 50 samples are used here. A larger sample of 500 data points is used here to test whether higher sample sizes are significantly beneficial to the control in a realistic setting. The results are benchmarked against the case with randomly selected stop positions without the application of control and a case where the SO problem is solved directly using the simulation with a termination time limit of 10 min allowing the solver to reach an optimal solution. This solution may not be reached with the time duration available for real-time control but is considered here to find the best control performance and compare the surrogate-based solution to it. Surrogate-based optimization is carried out under five different random seed values for each sample size to reduce any bias in results due to the training sample.

The studied application considers a total simulation duration of 900 s with 10 approaching vehicles with entry times at 30, 60, 90, 120, 250, 270, 300, 330, 520, and 550 s, and corresponding stop duration of 60, 120, 90, 60, 60, 120, 90, 60, 90, and 120 s, respectively. The MPC trigger range is set to 150 s(i.e. the control problem arises if an approaching vehicle is expected to enter the stop segment within the next 150 s), and the MPC's prediction horizon is maintained at 600 s. The length of the stop segment is 450 m and a traffic signal is installed at the downstream end. The segment has 2 lanes with the service vehicles using one of the lanes for stopping to complete their assigned operations. The FD parameters and the traffic signal cycle times are defined in Appendix 2. The simulation starts with the signal's red cycle in the first 20 s. The initial density is set at 0.02 veh/m throughout the length of the segment. The demand and supply remain equal to q_m for the entire simulation duration, defined over 10- s intervals. The duration of the time step used for computing the objective function (10) is computed using 10- s steps according to the simulation framework described in Section 3.1.

To analyze the control under varying detour penalty weights, five sets of weights are considered. All the weights are set to 0 in the first case. The remaining four cases are created by randomly picking the weights from uniform distributions with bounds $[10^{-4}, 10^{-3}]$, $[10^{-3}, 10^{-2}]$, $[10^{-2}, 10^{-1}]$, and $[10^{-1}, 1.0]$, respectively. A single random seed value is used for picking the random numbers. This is equivalent to randomly selecting a set of detour penalty weights for one of the ranges and scaling them according to the respective range in any case. This is done to keep the same relative magnitude of weights for the different vehicles in the four cases. The values for the range $[10^{-1}, 1.0]$ are 0.48, 0.75, 0.10, 0.37, 0.23, 0.18, 0.27, 0.41, 0.46, and 0.58 for the ten vehicles, respectively. We do not consider any constraints on the stop positions apart from the stability constraints defined in Section 3.2.

The stop positions in the no-control case are selected randomly from the set of feasible stop positions. The objective function values for the no-control scenario are presented in Table 5. The table also displays the values of the three components of the objective function (10). The detour penalty is computed by treating the original stop positions as the initially desired stop positions X_D, therefore, resulting in a value of 0 in the no-control case. The improvement achieved with the control using the GPR model-based surrogate modelling approach and the direct simulation-based approach is also presented in Table 5 for different sets of detour penalty weights. For the GPR model-based approach with both sample sizes, the average values over the five runs with different random seeds are presented in the table. Percentage improvements are presented for the overall objective, outflow, and spillback penalty while absolute improvement is presented for the detour penalty (since the value in the no-control case is 0). The negative signs indicate a worsening of the component compared to

Table 5. Objective function values for the different cases in the numerical example.

Control scenario	Detour penalty weight range	Objective	Outflow	Spillback penalty	Detour penalty
No-control	–	44.68	28.11	72.78	0.0
Simulation-based		20.65%	2.38%	11.74%	0.0
GPR with 50 samples	0.0	17.94%	1.74%	10.33%	0.0
GPR with 500 samples		17.08%	1.22%	10%	0.0
Simulation-based		19.15%	2.39%	11.47%	−0.43
GPR with 50 samples	$[10^{-4}, 10^{-3}]$	17.27%	1.52%	10.41%	−0.30
GPR with 500 samples		14.96%	0.58%	9.37%	−0.31
Simulation-based		12.68%	1.71%	10.22%	−2.26
GPR with 50 samples	$[10^{-3}, 10^{-2}]$	5.55%	−0.17%	5.74%	−1.66
GPR with 500 samples		5.72%	−0.09%	5.73%	−1.60
Simulation-based		0.58%	0.53%	2.34%	−1.60
GPR with 50 samples	$[10^{-2}, 10^{-1}]$	−1.94%	−0.02%	1.49%	−1.95
GPR with 500 samples		−1.83%	−0.26%	1.25%	−1.67
Simulation-based		0.00%	0.00%	0.00%	0.0
GPR with 50 samples	$[10^{-1}, 1.0]$	−24.22%	−0.12%	0.15%	−10.9
GPR with 500 samples		−2.75%	−0.06%	−0.08%	−1.16

the no-control case which is not desirable. In general, the spillback and detour penalty is supposed to be minimized while the total outflow is supposed to be maximized which results in the overall minimization of the objective function. The stop positions are presented in Table A3 in Appendix 3.

With a 0 detour penalty, the simulation-based solution is observed to produce a significant improvement of 20.65% in the objective value. The majority of the improvement comes from the spillback penalty terms while the outflow is also observed to improve. The corresponding improvement using the surrogate model is smaller but still significant at 17.94% and 17.08% with 50 and 500 samples, respectively. The improvement in the objective function value decreases with an increase in the value of the detour penalty weights for all controllers. The detour penalty itself decreases for all controllers after a point. This is because the increasing cost of detours offsets any improvement in traffic and eventually forces the controller to allow vehicles to stop at their desired positions. A detour in the cases with a high detour cost would only result in a large penalty resulting in a lower objective value than the no-control case. The simulation-based solutions are able to avoid this in all cases with a minimum improvement of 0.0% in the case with the highest detour costs. The GPR-based solutions do face the issue of a deterioration of the objective under control at large values of detour penalty weights. This is especially true with 50 samples which produces a worse solution than the no-control case in the two cases with the highest detour penalty weights. The controller with 500 samples also causes a deterioration in the two cases but it is smaller than the controller with 50 samples. This makes sense since a larger detour penalty creates a more complex relationship between the stop positions and objective function as it counters the improvement from the other terms in the objective. More samples result in better learning of the relationship and hence better results. Note, that the best performance of the controllers for the case with the highest detour costs are 7.27% with 50 samples, and 0.0% with 500 samples, respectively which shows that the models are indeed able to learn the relationships between the decision variables and the objectives better than what the averages convey although the performance is not always consistent. In this study, we utilize the parallel computing toolbox in MATLAB (The Mathworks Inc. 2021b) to generate the training samples with a pool of 6 parallel workers since the sampled points are independent of each other. The average computation time over the five cases with different detour penalties and over different seed values using the surrogate-based controllers with 50 and 500 samples are 6.55 and 21.26 s, respectively.

Figure 11 presents the plot of the distance between the optimal stop position and the desired stop position for the 10 vehicles in the no-control case and the control cases under different detour penalty weights using GPR with 500 samples.

The *x*-axis represents the vehicle index in the order of entry into the segment and the *y*-axis represents the absolute distance between the optimal and desired stop positions. It can be observed that

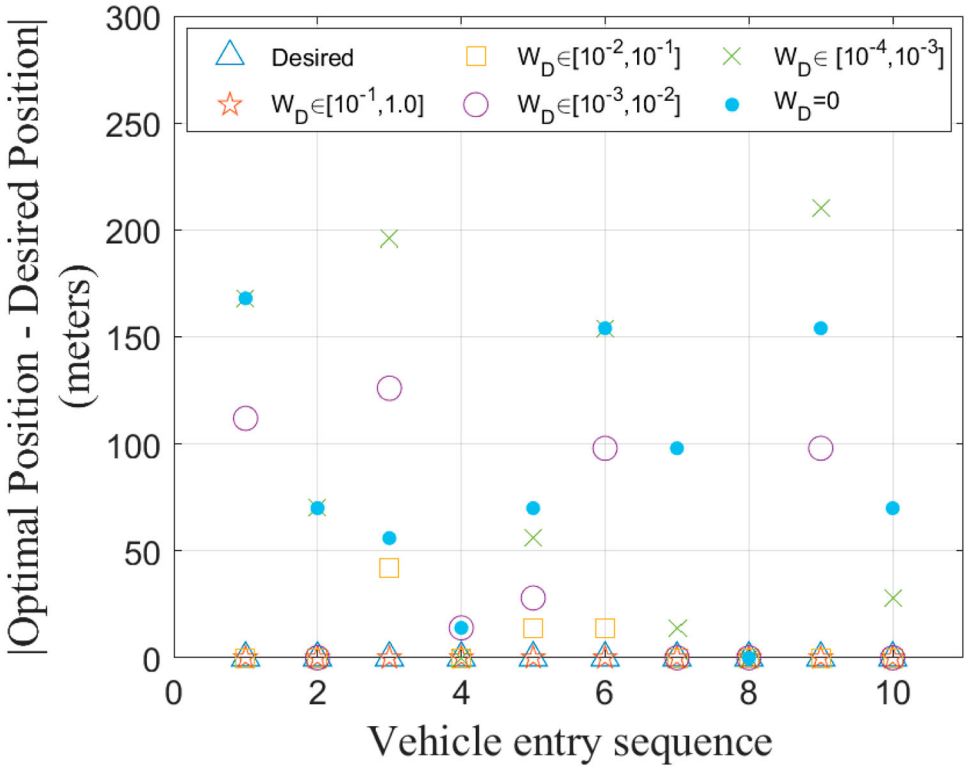


Figure 11. Absolute distance of optimal stop position from desired stop position under different ranges of detour penalty weights.

in most cases the distance between the desired positions and the optimal stop positions increases as the magnitude of the penalty weight decreases since the solver is given more flexibility to search for solutions with a large detour. In some cases, such as in Vehicle 3, this trend may not be exact as some solutions may need to be adjusted according to the position of other vehicles or there may be a range of stop positions for a vehicle that result in the same overall outcome. In all the cases, the stop positions with the largest detour penalty are the same as the desired position without control.

Figure 12 shows the space-time-density diagrams for the simulated traffic for both the no-control case and the control cases using simulation and the GPR model with 500 samples, with 0 detour penalty weights. Figure 12(b,c) show that the controller strategically arranges vehicle stops to overlap congestion waves, resulting in reduced spillback duration at the upstream end. In Figure 12b, Vehicle 1 stops further downstream compared to the no-control case and Vehicles 2, 3, and 4 make stops in the direction of propagation of the congestion wave of Vehicle 1 (highlighted with a dashed rectangle around the vehicle stops). The delay in Vehicle 1's stop allows future vehicles to rearrange their stops so the congestion waves may overlap, preventing additional congestion caused by their stops. Notably, the controlled case exhibits a more significant congestion wave from the fourth red cycle of the traffic signal, leading to a longer upstream congestion duration. This congestion was not observed in the no-control case as the traffic approaching the signal was blocked by vehicle stops, causing earlier congestion. By preventing early additional congestion, the controlled case reduces the cumulative time vehicles are blocked upstream (accounted for in the spillback penalty) resulting in a lower objective function value. Both, the simulation-based and surrogate-based controllers achieve an improvement by arranging vehicle stops to overlap congestion waves and delay congestion, albeit at different positions within the segment. For instance, in the discussed scenario, the simulation-based solution stops Vehicle 6 more upstream at 84 m (see the stop made by the first vehicle to stop in the dotted box

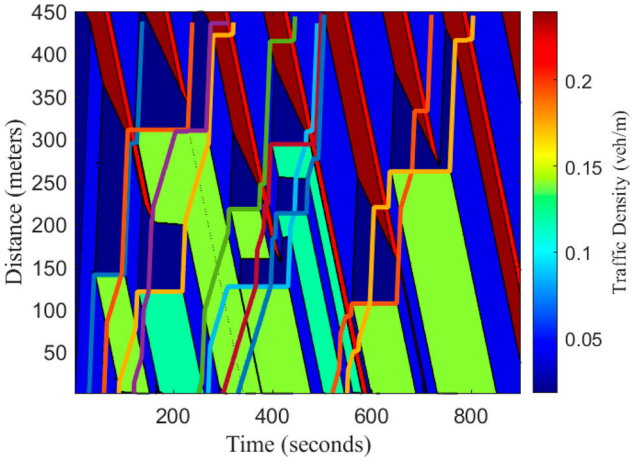


Figure 12. Traffic evolution considering (a) randomly selected stop positions, (b) optimally selected stop positions based on simulation, and (c) optimally selected stop positions based on the GPR model with 500 samples. The colour bar represents traffic density in veh/m.

drawn in Figure 12(b)), while the surrogate-based solution stops it at 280 m (see the stop made by the first vehicle to enter the dotted box drawn on Figure 12(c)). In both cases, Vehicles 7 and 8 (the second and third vehicles to enter the dotted box in both figures) stop upstream of Vehicle 6 at 56 m, 28 m, and 182 m, 210 m, respectively. As such, both solutions are valid and high-performing in this context.

6. Conclusion

This work focuses on the problem of optimal control of stop positions of vehicles intending to make a stop on a given stretch of the road to minimize congestion and maximize the total outflow. An MPC framework is developed to perform real-time control using an LWR simulation solved by means of an extended Lax–Hopf approach to reproduce stopping vehicles as moving bottlenecks. This work employs surrogate modelling to avoid computationally demanding SO strategies (that typically involve meta-heuristic optimization such as GA). Alternative surrogates are explored to replace the simulation in the solution approach with a model that closely captures the relationship between the simulation inputs and the objective function. The analyses showed that the number of simulation runs needed to train such models to an acceptable level is lower than the total simulations run within the GA to reach comparable results by reaching convergence criteria. They are, therefore, a time-efficient approach for the optimal control decision and allow for real-time control. The performed tests also highlight that GPR outperforms all the other techniques in terms of computational time and accuracy. A numerical study demonstrates that the performance of controls derived through surrogate-based optimization is equal to a simulation-based solution while requiring much less time.

The proposed method can be used to handle congestion from both downstream links as well as caused by traffic entering from the upstream end by accounting for the upstream and downstream boundary conditions in the traffic model. It can be used in both peak and non-peak hours and is especially useful for peak-hour traffic when temporary bottlenecks are more likely to cause severe congestion. However, the optimization may not be very effective on very short links since the traffic dynamics are mainly influenced by the traffic signals rather than the temporary bottlenecks. The proposed method can also be applied to the problem of optimal speed regulation of connected and autonomous cars to reduce traffic congestion and increase throughput. The method can also address bus stop locations for areas with a high density of buses. The use of the Lax–Hopf method makes it more attractive than CTM-based schemes in the literature due to the known computational benefits offered by the Lax–Hopf method, which is grid-free in nature and has a semi-analytical mathematical form. Besides, the inclusion of capacity drop in the simulation framework is another possible extension of the work which will make it more reliable for real-world control. This is possible by incorporating the work on Lax–Hopf solutions to the LWR model with bounded acceleration by Qian et al. (2019).

Lastly, this work focuses on optimizing short stops at the curbside in scenarios where no dedicated parking infrastructure is available. It does not address specific dynamics associated with the presence of parking infrastructure that is instead, investigated in more detail in Simoni (2024). The cruising behaviour associated with parking is also not modelled in this study, as we assume the network operator provides prior information on available spots. However, low compliance scenarios where drivers may cruise to find parking could be a subject for future work. Additionally, while this study assumes fixed stop durations for all vehicles, future work could explore stochastic variations in stop times, to ensure robust decision-making.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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References

Adacher, L. 2012. "A Global Optimization Approach to Solve the Traffic Signal Synchronization Problem." *Procedia-Social and Behavioral Sciences* 54:1270–1277. https://doi.org/10.1016/j.sbspro.2012.09.841.

Al-Turjman, Fadi, and Arman Malekloo. 2019. "Smart Parking in IoT-enabled Cities: A Survey." *Sustainable Cities and Society* 49:101608. https://doi.org/10.1016/j.scs.2019.101608.

Alho, André Romano, João de Abreu e Silva, Jorge Pinho de Sousa, and Edgar Blanco. 2018. "Improving Mobility by Optimizing the Number, Location and Usage of Loading/unloading Bays for Urban Freight Vehicles." *Transportation Research Part D: Transport and Environment* 61:3–18. https://doi.org/10.1016/j.trd.2017.05.014.

Aubin, J. P., A. M. Bayen, and P. Saint-Pierre. 2008. "Dirichlet Problems for Some Hamilton–Jacobi Equations with Inequality Constraints." *SIAM Journal on Control and Optimization* 47 (5): 2348–2380. https://doi.org/10.1137/060659569.

Breiman, Leo. 1996. "Bagging Predictors." *Machine Learning* 24 (2): 123–140.

Burns, Aaron J., Jeremy J. Michalek, and Constantine Samaras. 2024. "Estimating the Potential for Optimized Curb Management to Reduce Delivery Vehicle Double Parking, Traffic Congestion and Energy Consumption." *Transportation Research Part E: Logistics and Transportation Review*187:103574. https://doi.org/10.1016/j.tre.2024.103574.

Chen, Xiqun Michael, Chenfeng Xiong, Xiang He, Zheng Zhu, and Lei Zhang. 2016. "Time-of-Day Vehicle Mileage Fees for Congestion Mitigation and Revenue Generation: A Simulation-Based Optimization Method and Its Real-World Application." *Transportation Research Part C: Emerging Technologies*63:71–95. https://doi.org/10.1016/j.trc.2015.12.001.

Chen, Xiqun, Lei Zhang, Xiang He, Chenfeng Xiong, and Zhiheng Li. 2014. "Surrogate-Based Optimization of Expensive-to-Evaluate Objective for Optimal Highway Toll Charges in Transportation Network." *Computer-Aided Civil and Infrastructure Engineering* 29 (5): 359–381. https://doi.org/10.1111/mice.2014.29.issue-5.

Claudel, Christian G., and Alexandre M. Bayen. 2010a. "Lax–Hopf Based Incorporation of Internal Boundary Conditions into Hamilton–Jacobi Equation. Part I: Theory." *IEEE Transactions on Automatic Control* 55 (5): 1142–1157. https://doi.org/10.1109/TAC.2010.2041976.

Claudel, Christian G., and Alexandre M. Bayen. 2010b. "Lax–Hopf Based Incorporation of Internal Boundary Conditions into Hamilton–Jacobi Equation. Part II: Computational Methods." *IEEE Transactions on Automatic Control* 55 (5): 1158–1174. https://doi.org/10.1109/TAC.2010.2045439.

Comi, Antonio, Berta Buttarazzi, Massimiliano M. Schiraldi, Rosy Innarella, Martina Varisco, and Luca Rosati. 2017. "DynaLOAD: A Simulation Framework for Planning, Managing and Controlling Urban Delivery Bays." *Transportation Research Procedia* 22:335–344. https://doi.org/10.1016/j.trpro.2017.03.049.

Courant, Richard, Kurt Friedrichs, and Hans Lewy. 1967. "On the Partial Difference Equations of Mathematical Physics." *IBM Journal of Research and Development* 11 (2): 215–234. https://doi.org/10.1147/rd.112.0215.

Daganzo, Carlos F. 1994. "The Cell Transmission Model: A Dynamic Representation of Highway Traffic Consistent with the Hydrodynamic Theory." *Transportation Research Part B: Methodological* 28 (4): 269–287. https://doi.org/10.1016/0191-2615(94)90002-7.

Daganzo, Carlos F. 1995. "The Cell Transmission Model, Part II: Network Traffic." *Transportation Research Part B: Methodological* 29 (2): 79–93. https://doi.org/10.1016/0191-2615(94)00022-R.

do Amaral, João Victor Soares, José Arnaldo Barra Montevechi, Rafael de Carvalho Miranda, and Wilson Trigueiro de Sousa Jr.. 2022. "Metamodel-Based Simulation Optimization: A Systematic Literature Review." *Simulation Modelling Practice and Theory* 114:102403. https://doi.org/10.1016/j.simpat.2021.102403.

Hassannayebi, Erfan, Arman Sajedinejad, Ali Kardannia, Masoud Shakibayifar, Hossein Jafari, and Ehsan Mansouri. 2021. "Simulation-Optimization Framework for Train Rescheduling in Rapid Rail Transit." *Transportmetrica B: Transport Dynamics* 9 (1): 343–375.

Hopf, Eberhard. 1969. "On the Right Weak Solution of the Cauchy Problem for a Quasilinear Equation of First Order." *Journal of Mathematics and Mechanics* 19 (6): 483–487.

Jain, Milan, Vinay C. Amatya, Amelia Bleeker, Soumya Vasisht, John T. Feo, and Katherine E. Wolf. 2024. "Predicting Curb Side Parking Availability for Commercial Vehicle Loading Zones." *International Journal of Intelligent Transportation Systems Research* 22 (3): 614–628. https://doi.org/10.1007/s13177-024-00420-5.

Joachims, Thorsten. 2005. "A Support Vector Method for Multivariate Performance Measures." In *Proceedings of the 22nd International Conference on Machine Learning*, 377–384.

Kouvaritakis, Basil, and Mark Cannon. 2016. *Model Predictive Control*. Vol. 38. Switzerland: Springer International Publishing.

Lax, Peter D. 1957. "Hyperbolic Systems of Conservation Laws II." *Communications on Pure and Applied Mathematics* 10 (4): 537–566. https://doi.org/10.1002/cpa.v10:4.

Lemaréchal, Claude. 2012. "Cauchy and the Gradient Method." *Doc Math Extra* 251 (254): 10.

Letnik, Tomislav, Alessandro Farina, Matej Mencinger, Marino Lupi, and Stane Božičnik. 2018. "Dynamic Management of Loading Bays for Energy Efficient Urban Freight Deliveries." *Energy*159:916–928. https://doi.org/10.1016/j.energy.2018.06.125.

Liang, Yongsheng, Zhigang Ren, Lin Wang, Hanqing Liu, and Wenhao Du. 2021. "Surrogate-Assisted Cooperative Signal Optimization for Large-Scale Traffic Networks." *Knowledge-Based Systems*234:107542. https://doi.org/10.1016/j.knosys.2021.107542.

Liard, Thibault, Raphael Stern, and Maria Laura Delle Monache. 2023. "A PDE-ODE Model for Traffic Control with Autonomous Vehicles ." *Networks and Heterogeneous Media* 18 (3): 1190–1206.

Lighthill, Michael James, and Gerald Beresford Whitham. 1955. "On Kinematic Waves II. A Theory of Traffic Flow on Long Crowded Roads." *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences* 229 (1178): 317–345.

Lim, Jisoon, and Neda Masoud. 2024. "Dynamic Usage Allocation and Pricing for Curbside Space Operation." *Transportation Science* 58 (6): 1252–1276.

Liu, Jiachao, Wei Ma, and Sean Qian. 2023. "Optimal Curbside Pricing for Managing Ride-Hailing Pick-Ups and Drop-Offs." *Transportation Research Part C: Emerging Technologies* 146:103960. https://doi.org/10.1016/j.trc.2022.103960.

Liu, Jiachao, and Sean Qian. 2024. "Modeling Multimodal Curbside Usage in Dynamic Networks." *Transportation Science* 58 (6): 1277–1299. https://doi.org/10.1287/trsc.2024.0522.

Liu, Xiaohui, Sean Qian, Hock-Hai Teo, and Wei Ma. 2024. "Estimating and Mitigating the Congestion Effect of Curbside Pick-Ups and Drop-Offs: A Causal Inference Approach." *Transportation Science* 58 (2): 355–376. https://doi.org/10.1287/trsc.2022.0195.

Marsden, Greg, Iain Docherty, and Robyn Dowling. 2020. "Parking Futures: Curbside Management in the Era of 'new Mobility' Services in British and Australian Cities." *Land Use Policy* 91:104012. https://doi.org/10.1016/j.landusepol.2019.05.031.

The Mathworks Inc. 2021a. *MATLAB Global Optimization Toolbox Release*. Natick, Massachusetts, United States: MathWorks Inc.

The Mathworks Inc.. 2021b. *MATLAB Parallel Computing Toolbox Release*. Natick, Massachusetts, United States: MathWorks Inc.

Maxner, Thomas, Andisheh Ranjbari, Chase P. Dowling, and Şeyma Güneş. 2023. "Simulation-Based Analysis of Different Curb Space Type Allocations on Curb Performance." *Transportmetrica B: Transport Dynamics* 11 (1): 1384–1405.

Mazare, P. E., A. Dehwah, C. G. Claudel, and A. M. Bayen. 2011. "Analytical and Grid-Free Solutions to the Lighthill–Whitham–Richards Traffic Flow Model." *Transportation Research Part B: Methodological*45 (10): 1727–1748. https://doi.org/10.1016/j.trb.2011.07.004.

McLeod, Fraser, and Tom Cherrett. 2011. "Loading Bay Booking and Control for Urban Freight." *International Journal of Logistics Research and Applications* 14 (6): 385–397. https://doi.org/10.1080/13675567.2011.641525.

Mor, A., M. G. Speranza, and J. M. Viegas. 2020. "Efficient Loading and Unloading Operations Via a Booking System." *Transportation Research Part E: Logistics and Transportation Review* 141:102040. https://doi.org/10.1016/j.tre.2020.102040.

Muriel, Juan E., Lele Zhang, Jan C. Fransoo, and Juan G. Villegas. 2024. "A Reinforcement Learning Framework for Improving Parking Decisions in Last-Mile Delivery." *Transportmetrica B: Transport Dynamics* 12 (1): 2337216.

Osorio, Carolina. 2019. "High-Dimensional Offline Origin-Destination (OD) Demand Calibration for Stochastic Traffic Simulators of Large-Scale Road Networks." *Transportation Research Part B: Methodological* 124:18–43. https://doi.org/10.1016/j.trb.2019.01.005.

Osorio, Carolina, and Michel Bierlaire. 2013. "A Simulation-Based Optimization Framework for Urban Transportation Problems." *Operations Research* 61 (6): 1333–1345. https://doi.org/10.1287/opre.2013.1226.

Osorio, Carolina, and Linsen Chong. 2012. "An Efficient Simulation-Based Optimization Algorithm for Large-Scale Transportation Problems." In *Proceedings of the 2012 Winter Simulation Conference (WSC)*, 1–11.

Piacentini, Giulia, Antonella Ferrara, Ioannis Papamichail, and Markos Papageorgiou. 2019. "Highway Traffic Control with Moving Bottlenecks of Connected and Automated Vehicles for Travel Time Reduction." In *2019 IEEE 58th Conference on Decision and Control (CDC)*, 3140–3145. IEEE.

Piacentini, Giulia, Paola Goatin, and Antonella Ferrara. 2018. "Traffic Control Via Moving Bottleneck of Coordinated Vehicles." *IFAC-PapersOnLine* 51 (9): 13–18. https://doi.org/10.1016/j.ifacol.2018.07.003.

Piacentini, G., C. Pasquale, S. Sacone, S. Siri, and A. Ferrara. 2019. "Multiple Moving Bottlenecks for Traffic Control in Freeway Systems." In *2019 18th European Control Conference (ECC)*, 3662–3667.

Pinto, Roberto, Alexandra Lagorio, and Ruggero Golini. 2019. "The Location and Sizing of Urban Freight Loading/unloading Lay-By Areas." *International Journal of Production Research* 57 (1): 83–99. https://doi.org/10.1080/00207543.2018.1461269.

Qian, Kun, Guillaume Costeseque, Edward S. Canepa, and Christian G. Claudel. 2019. "Optimization-Based Queue Length Estimation on Arterial Networks Using a Macroscopic Traffic Model Enforcing Vehicles Bounded Acceleration." ⟨hal-02175497v2⟩.

Ranjbari, Andisheh, JosÅ Luis Machado-León, Giacomo Dalla Chiara, Don MacKenzie, and Anne Goodchild. 2021. "Testing Curbside Management Strategies to Mitigate the Impacts of Ridesourcing Services on Traffic." *Transportation Research Record* 2675 (2): 219–232. https://doi.org/10.1177/0361198120957314.

Rasmussen, Carl Edward. 2003. "Gaussian Processes in Machine Learning." In *Summer School on Machine Learning*, 63–71. Berlin, Heidelberg: Springer.

Richards, Paul I. 1956. "Shock Waves on the Highway." *Operations Research* 4 (1): 42–51. https://doi.org/10.1287/opre.4.1.42.

Rizwan, Patan, K. Suresh, and M. Rajasekhara Babu. 2016. "Real-Time Smart Traffic Management System for Smart Cities by Using Internet of Things and Big Data." In *2016 International Conference on Emerging Technological Trends (ICETT)*, 1–7. IEEE.

Roca-Riu, Mireia, Jin Cao, Igor Dakic, and Monica Menendez. 2017. "Designing Dynamic Delivery Parking Spots in Urban Areas to Reduce Traffic Disruptions." *Journal of Advanced Transportation*2017:1–15. https://doi.org/10.1155/2017/6296720.

Simoni, Michele D. 2024. "Parking Guidance and Geofencing for Last-Mile Delivery Operations." *IEEE Transactions on Intelligent Transportation Systems* 25 (8): 9091–9102.

Simoni, Michele D., and Christian G. Claudel. 2017. "A Fast Simulation Algorithm for Multiple Moving Bottlenecks and Applications in Urban Freight Traffic Management." *Transportation Research Part B: Methodological* 104:238–255. https://doi.org/10.1016/j.trb.2017.06.010.

Simoni, Michele D., and Christian G. Claudel. 2018. "A Simulation Framework for Modeling Urban Freight Operations Impacts on Traffic Networks." *Simulation Modelling Practice and Theory* 86:36–54. https://doi.org/10.1016/j.simpat.2018.05.001.

Stueger, Philipp N., Fabian Fehn, and Klaus Bogenberger. 2023. "Minimizing the Effects of Urban Mobility-On-Demand Pick-Up and Drop-off Stops: A Microscopic Simulation Approach." *Transportation Research Record* 2677 (1): 814–828. https://doi.org/10.1177/03611981221101894.

Suratgar, Amir Abolfazl, Mohammad Bagher Tavakoli, and Abbas Hoseinabadi. 2005. "Modified Levenberg–Marquardt Method for Neural Networks Training." *World Academy of Science, Engineering and Technology* 6 (1): 46–48.

Tavassoli, Abbas, Yadollah Waghei, and Alireza Nazemi. 2022. "Comparison of Kriging and Artificial Neural Network Models for the Prediction of Spatial Data." *Journal of Statistical Computation and Simulation* 92 (2): 352–369. https://doi.org/10.1080/00949655.2021.1961140.

Ukkusuri, Satish V., Kaan Ozbay, Wilfredo F. Yushimito, Shri Iyer, Ender F. Morgul, and José Holguín-Veras. 2016. "Assessing the Impact of Urban Off-Hour Delivery Program Using City Scale Simulation Models." *EURO Journal on Transportation and Logistics* 5 (2): 205–230. https://doi.org/10.1007/s13676-015-0079-3.

Wang, Yiyun, Rongjie Yu, Shuhan Qiu, Jian Sun, and Haneen Farah. 2022. "Safety Performance Boundary Identification of Highly Automated Vehicles: A Surrogate Model-Based Gradient Descent Searching Approach." *IEEE Transactions on Intelligent Transportation Systems* 23 (12): 23809–23820. https://doi.org/10.1109/TITS.2022.3191088.

Xiong, Chenfeng, Zheng Zhu, Xiqun Chen, and Lei Zhang. 2018. "Optimal Travel Information Provision Strategies: An Agent-Based Approach under Uncertainty." *Transportmetrica B: Transport Dynamics* 6 (2): 129–150.

Xu, Zhengtian, and Xiaotong Sun. 2024. "Economic Analysis of On-Street Parking with Urban Delivery." *Transportation Science* 58 (6): 1300–1318. https://doi.org/10.1287/trsc.2024.0569.

Yang, Kaidi, Mireia Roca-Riu, and Mónica Menéndez. 2019. "An Auction-Based Approach for Prebooked Urban Logistics Facilities." *Omega*89:193–211. https://doi.org/10.1016/j.omega.2018.10.005.

Ye, Qiming, Simon M. Stebbins, Yuxiang Feng, Eduardo Candela, Marc Stettler, and Panagiotis Angeloudis. 2020. "Intelligent Management of On-Street Parking Provision for the Autonomous Vehicles Era." In *2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)*, 1–7. IEEE.

Zhang, Chao, Carolina Osorio, and Gunnar Flötteröd. 2017. "Efficient Calibration Techniques for Large-Scale Traffic Simulators." *Transportation Research Part B: Methodological* 97:214–239. https://doi.org/10.1016/j.trb.2016.12.005.

Zhang, He, Huajun Zhou, Jian Sun, and Ye Tian. 2022. "Risk Assessment of Highly Automated Vehicles with Naturalistic Driving Data: A Surrogate-Based Optimization Method." In *2022 IEEE Intelligent Vehicles Symposium (IV)*, 580–585. IEEE.

Appendices

Appendix 1. Summary of techniques

Here we provide a brief description of the modelling techniques used in this study in context to our problem.

(1) Linear Regression.

The linear regression model can be represented as

$$Y = \beta_0 + X^T \beta_1,$$

where Y represents the objective function of the optimization problem presented in (10), and X represents the inputs with corresponding stop positions of the approaching vehicles in case of the *local* model and also the simulation parameters in case of the *global* model. β_0 and β_1 are the fitted model coefficients where β_1 represents the impact of the input variables on the value of the objective function and β_0 is the intercept.

(2) Polynomial Regression

The polynomial regression model can be represented as

$$Y = \beta_0 + X^T \beta_1 + X^T B X,$$

where X and Y are the same as for the linear regression model. The third term in the above equation represents the effect of the interaction between inputs on the objective function value where B is a matrix consisting of the coefficients of the second-order terms. β_0, β_1, and B are fitted from the data.

(3) Regression Tree Ensemble

A regression tree is a non-parametric regression model consisting of a binary tree obtained by recursively splitting the data on the predictor values. Here, the predictors denote the input variables X as described above. The predictor and the corresponding split value at each node are selected such that the variance of the output (Y) around its mean value on the child nodes is minimized. The variables and split values form the parameters of the model. The output of a regression tree is the average output value in the end nodes of the tree. A regression tree ensemble is a weighted combination of several such trees. Each tree is trained on a different set of data and the final output is the weighted mean of the predictions from all individual trees. In this work, we use the bagging technique (Breiman 1996) for training the trees which involves producing a bootstrap using the training data. The fitrensemble() method of MATLAB is used to implement the same.

(4) Neural Network

A feedforward neural network or simply neural network consists of an input layer with as many nodes as the number of inputs X, an output layer with one node for the output Y, and an arbitrary number of hidden layers or intermediate layers with an arbitrary number of nodes per layer. Every node in each layer is connected to every node in the next layer and a weight is associated with each such connection. Additionally, a bias is associated with all the nodes. The neural network is trained to find the optimal values of these weights and biases to minimize the error in the final output. Nodes in each layer other than the input layer utilize a *transfer function* to process the received values and send the resulting value to the next layer. In this work, we use the fitnet() function of MATLAB which corresponds to the following hyperbolic tangent sigmoid transfer function:

$$o = \frac{2}{1 + \exp(-2i)} - 1$$

where i denotes the input to the node and o denotes the output. The network is trained using the Levenberg–Marquardt optimization method (Suratgar, Tavakoli, and Hoseinabadi 2005). In this work, we use two hidden layers with 24 nodes in each hidden layer in the case of the *global* models which have a large number of inputs (decision variables and simulation parameters) and 4 nodes in the case of the *local* models with a small number of inputs (only the decision variables).

(5) Radial basis neural network

A radial basis neural network is a variant of the neural network which has one hidden layer with the *transfer function* expressed as the following radial basis function:

$$o = \exp(-i^2).$$

In this work, we utilize the newrb() function of MATLAB to implement a radial basis neural network. As input we set a target mean squared error equal to 0.01, and a spread of the radial basis function equal to 2000. In general, a larger spread means that more neurons will be added to the hidden layer to fit a fast-changing function. The maximum number of neurons in the hidden layer is kept as 4 in the case of *local* models and 24 in the case of *global* models.

(6) Gaussian Process Regression (Kriging)

A Gaussian process regression or kriging model (Rasmussen 2003) is a non-parametric kernel-based probabilistic model of the form:

$$Y = h(x)^T \beta + f(x),$$

where $f(x) \sim GP(0, k(x, x^T))$ which is a zero mean Gaussian Process with a covariance function $k(x, x^T)$. The variable $h(x)$ is a set of basis functions that transform the original inputs to a new features vector, and β is a coefficient vector. Here, X and Y are the same as in the linear regression model. The functional forms of $h(x)$ and $f(x)$ are chosen in advance while the coefficients β and the hyper-parameters of the kernel function are fitted using the supplied data. In this work, we use a constant basis function $h(x)$ and a squared exponential kernel function of the form

$$k(x, x') = \theta_0^2 \exp\left(-\frac{(x - x')^2}{2\theta_1^2}\right)$$

where θ_0 and θ_1 are hyper-parameters of the kernel function. The given function selections are the default settings of the fitrgp() function of MATLAB which is utilized in this study to train a kriging model on the simulated data.

Appendix 2. Sub-scenarios description

Here we describe the sub-scenarios employed for testing the surrogate models' performance in Section 4.3. Notations for the scenarios are the same as in Table 2 that is C1, C2, C3, C4, and C5. The goal is to create a diverse set of scenarios to represent various situations that can be encountered when solving the SO problem over different control times. A total of

12 sub-scenarios are considered for the scenarios with no on-segment vehicles that is C1 and C3, and an additional sub-scenario is considered for scenarios with on-segment vehicles which allows varying the parameters associated with these vehicles. The first 12 sub-scenarios are common between C1, C2, and C3 as they do not involve on-segment vehicles. The same is also true for the first 12 sub-scenarios for C4 and C5. Certain simulation parameters which do not change with the time of solving the SO problem include the spatial limits of the stop segment, the FD parameters, and traffic signal cycle times. The length of the stop segment is kept at 450 m with a traffic signal installed at the downstream end. The total signal cycle time and the red-green splits for the traffic signal are set as 120 s and 48 s, respectively. The FD parameters are set as $v_f = 14$ m/sec, $\rho_c = 0.04$ veh/m, $\rho_m = 0.24$ veh/m, $w_c = 2.8$ m/sec, and $q_m = 0.56$ veh/sec for a 2-lane road with one of the lanes utilized for the pick-up/drop-off operations. Apart from these, all other simulation parameters may change depending on the time of solving the SO problem. These can be divided into the following categories- (1) traffic flow related which includes the initial density, demand, and supply for the stop segment, and the traffic signal red cycle start time, (2) on-segment vehicles-related which include their start positions, stop positions, and stop durations, and (3) approaching vehicles-related which include their stop durations and entry times into the stop segment. A prediction horizon of 600 s is considered for the analysis which is split into 6 steps of 100 s each such that demand and supply parameters are defined for each step resulting in 6 demand and 6 supply parameters. The length of the stop segment is discretized into 3 parts of 150 m each with an initial density defined for each part resulting in 3 initial density parameters. Each vehicle-related parameter contributes to as many simulation parameters as the number of vehicles. The various sub-scenarios are crafted by considering different combinations of values of these parameters. The value of the penalty term w_{SB} used for computing the objective function (10) is set as 0.1. The penalty vector W_D for detour from initial desired stop position is set to 0 in this case assuming that there is no desired stop positions considered.

Sub-scenario 1: The parameters for the first sub-scenario are described in Table A1. As mentioned above, the demand, supply, and initial density have 6, 6, and 3 parameters, respectively to account for their values over the discretized blocks. The demand and supply are defined for the first to the last time step from left to right. Similarly, the initial density is defined from the upstream to downstream end from left to right. The order of vehicles in the values for all the vehicle-related parameters is the same. Sub-scenario 1 can be considered as the default sub-scenario and all other sub-scenarios are created by varying one or more parameters from the values presented in Table A1.

Table A1. Parameter values for sub-scenario 1.

Parameter	Values
Traffic flow-related	
Demand (veh/sec)	$q_m, q_m, q_m, q_m, q_m, q_m$.
Supply (veh/sec)	$q_m, q_m, q_m, q_m, q_m, q_m$.
Initial density conditions (veh/m)	0.02, 0.02, 0.02.
Red signal start time (seconds)	20.
On-segment vehicles-related	
Start positions (meters)	70, 84, 112, 182.
Stop positions (meters)	112, 126, 196, 210.
Stop durations (seconds)	60, 60, 60, 60.
Approaching vehicles-related	
For C1, C2, and C3	
Entry times (seconds)	30, 60, 90, 120.
Stop durations (seconds)	60, 120, 90, 60.
For C4 and C5	
Entry times (seconds)	30, 50, 70, 90, 110, 130.
Stop durations (seconds)	60, 120, 90, 60, 120, 90.

The traffic flow-related parameters in Table A1 are common for all scenarios C1 to C5. The on-segment vehicle-related parameters in Table A1 are only valid for scenarios with on-segment vehicles, that is C2, C3, and C5. Since C2 and C5, only have 2 on-segment vehicles, values related to only the first two vehicles are relevant for them, whereas, for C3, all on-segment parameter values are relevant. Two sets of approaching vehicle-related parameters are defined for scenarios with different numbers of approaching vehicles.

Sub-scenarios 2–13: As mentioned above, the sub-scenarios 2 to 13 are defined by varying one or more parameter values from the default settings defined in Table A1. For each sub-scenario, Table A2 lists the parameters that are different from sub-scenario 1 and their corresponding values. Notice that sub-scenarios 2 to 8 only modify traffic flow-related parameters and are common between all the scenarios. Sub-scenarios 9 to 13 modify approaching and on-segment vehicle-related parameters and contain different variations for different sets of scenarios. Table A2 presents what each sub-scenario means for the corresponding scenario. Again, for sub-scenario 13, only the first two parameters from the on-segment vehicles-related parameters are relevant for C2 and C5 which only have 2 on-segment vehicles.

Table A2. Different parameter values in sub-scenarios 2–13 for various scenarios.

Sub-scenario	Parameter different from sub-scenario 1	Values
	For C1 to C5	
2	Red Signal Start Time	−30
3	Initial Density Conditions	0.08, 0.08, 0.08
4	Demand	$q_m/2, q_m/2, q_m, q_m, q_m, q_m$
5	Demand	$q_m, q_m, q_m/2, q_m/2, q_m, q_m$
6	Supply	$q_m/2, q_m/2, q_m, q_m, q_m, q_m$
7	Supply	$q_m, q_m, q_m/2, q_m/2, q_m, q_m$
8	Demand	$q_m, q_m/2, q_m, q_m/2, q_m, q_m$
	For C1 to C3	
9	Approaching entry times	10, 20, 30, 40
10	Approaching stop durations	120, 90, 120, 90
11	Demand	$q_m, q_m/2, q_m/2, q_m, q_m, q_m$
	Approaching entry times	10, 20, 30, 40
	Approaching stop durations	120, 90, 120, 90
12	Demand	$q_m, q_m/2, q_m/2, q_m, q_m, q_m$
	Approaching entry times	120, 130, 140, 150
	Approaching stop durations	120, 90, 120, 90
	For C4 and C5	
9	Approaching entry times	10, 20, 30, 40, 50, 60
10	Approaching stop durations	120, 90, 120, 90, 60, 120
11	Demand	$q_m, q_m/2, q_m/2, q_m, q_m, q_m$
	Approaching entry times	10, 20, 30, 40, 50, 60
	Approaching stop durations	120, 90, 120, 90, 60, 120
12	Demand	$q_m, q_m/2, q_m/2, q_m, q_m, q_m$
	Approaching entry times	100, 110, 120, 130, 140, 150
	Approaching stop durations	120, 90, 120, 90, 60, 120
	For C2, C3 and C5	
13	On-segment start positions	224, 252, 112, 182
	On-segment stop positions	266, 280, 140, 210
	On-segment stop durations	120, 60, 60, 120

Appendix 3. Additional numerical results

Here we present the optimal stop positions corresponding to the cases considered in Table 5.

Table A3. Optimal stop positions resulting in the best objective value out of the tested random seeds for the cases considered in the numerical example.

Control scenario	Detour penalty weight range	Stop positions of vehicles (in meters from the upstream boundary)
No-control	–	140, 308, 112, 308, 210, 126, 280, 210, 98, 252
Simulation-based		308, 224, 154, 126, 266, 84, 56, 28, 56, 280
GPR with 50 samples	0.0	308, 266, 238, 168, 280, 266, 210, 182, 266, 168
GPR with 500 samples		308, 238, 168, 294, 280, 280, 182, 210, 252, 182
Simulation-based		308, 224, 154, 168, 224, 112, 84, 56, 98, 56
GPR with 50 samples	$[10^{-4}, 10^{-3}]$	308, 266, 238, 294, 238, 266, 210, 210, 252, 168
GPR with 500 samples		308, 238, 308, 308, 266, 280, 266, 210, 308, 280
Simulation-based		308, 224, 154, 280, 224, 196, 294, 210, 168, 280
GPR with 50 samples	$[10^{-3}, 10^{-2}]$	294, 294, 238, 308, 266, 252, 280, 210, 224, 252
GPR with 500 samples		252, 308, 238, 294, 238, 224, 280, 210, 196, 252
Simulation-based		140, 308, 168, 280, 210, 126, 280, 210, 98, 252
GPR with 50 samples	$[10^{-2}, 10^{-1}]$	140, 308, 168, 308, 210, 140, 280, 210, 112, 252
GPR with 500 samples		140, 308, 154, 308, 224, 140, 280, 210, 98, 252
Simulation-based		140, 308, 112, 308, 210, 126, 280, 210, 98, 252
GPR with 50 samples	$[10^{-1}, 1.0]$	140, 308, 112, 308, 196, 126, 280, 210, 98, 252
GPR with 500 samples		140, 308, 112, 308, 210, 126, 280, 210, 98, 252